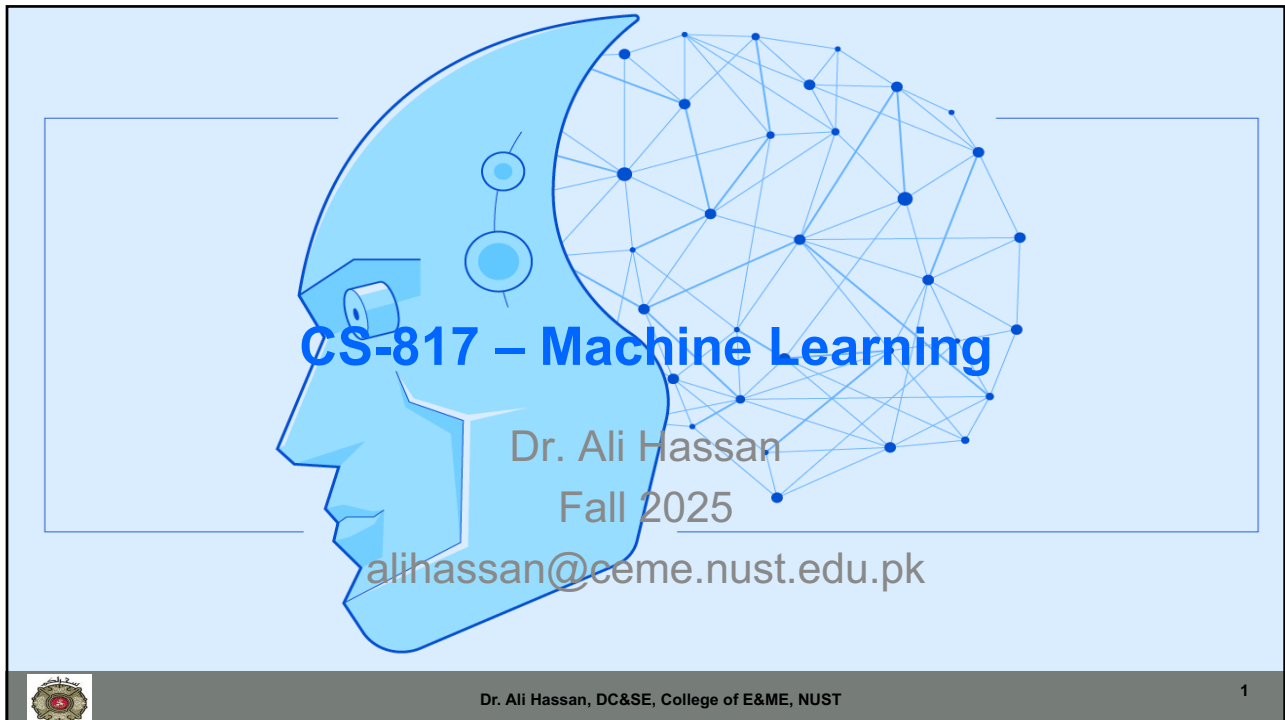


A blue-tinted graphic featuring a stylized human head profile on the left, composed of geometric shapes and circles. To the right of the head is a network diagram with numerous blue nodes connected by thin lines, resembling a neural network or a complex graph. The text "CS-817 – Machine Learning" is prominently displayed in the center in a bold, blue font. Below it, in a smaller, grey font, is "Dr. Ali Hassan", followed by "Fall 2025" and the email address "alihassan@ceme.nust.edu.pk". At the bottom left is a small circular logo, and at the bottom right is the number "1".

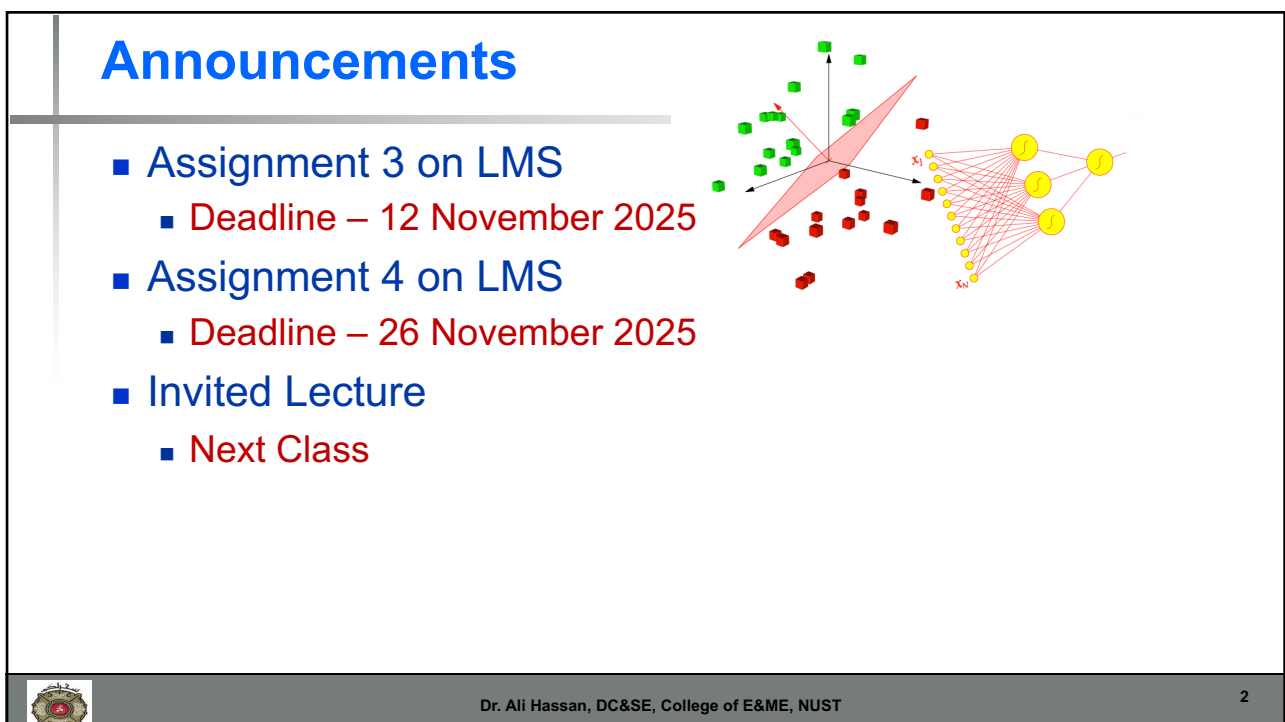
CS-817 – Machine Learning

Dr. Ali Hassan
Fall 2025
alihassan@ceme.nust.edu.pk

Dr. Ali Hassan, DC&SE, College of E&ME, NUST

1

1

The slide has a white background with a blue header "Announcements". Below the header is a list of three items, each preceded by a blue square bullet point. The first item is "Assignment 3 on LMS" with a sub-bullet "Deadline – 12 November 2025" in red. The second item is "Assignment 4 on LMS" with a sub-bullet "Deadline – 26 November 2025" in red. The third item is "Invited Lecture" with a sub-bullet "Next Class" in red. To the right of the text is a diagram showing a 3D coordinate system with green and red cubes, a red plane, and a neural network diagram with yellow nodes and red connections. At the bottom left is a small circular logo, and at the bottom right is the number "2".

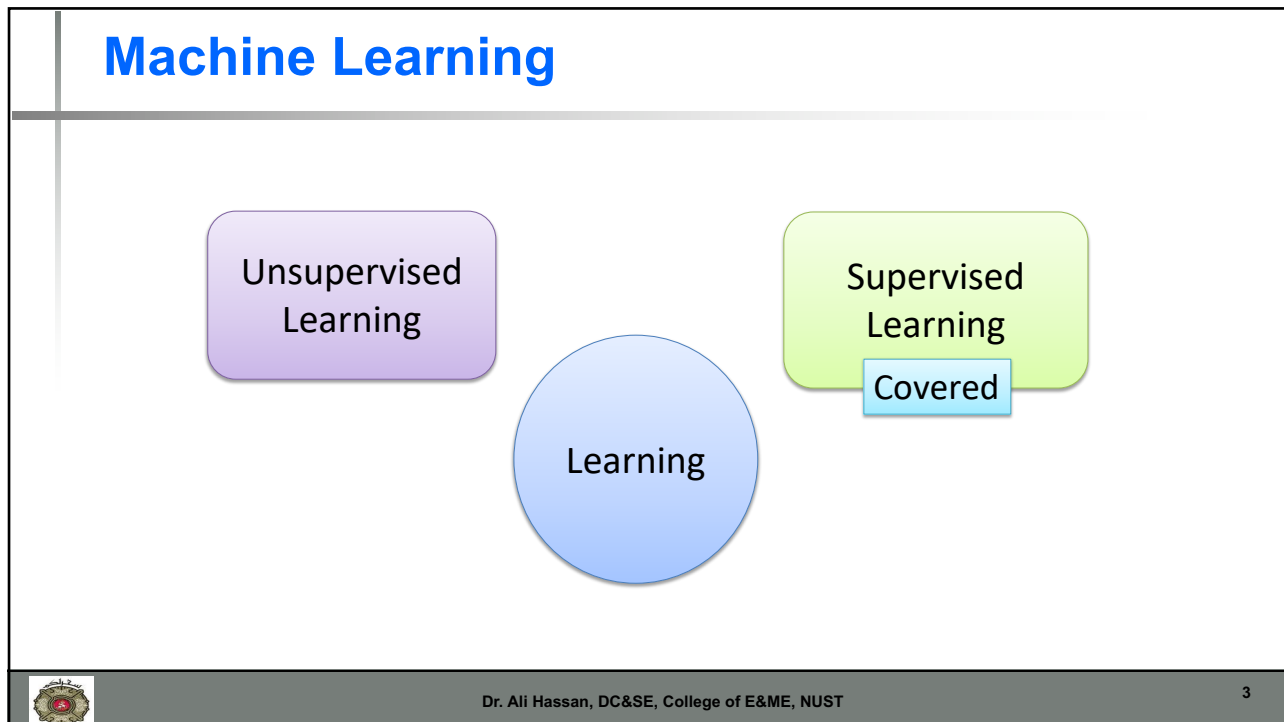
Announcements

- Assignment 3 on LMS
 - Deadline – 12 November 2025
- Assignment 4 on LMS
 - Deadline – 26 November 2025
- Invited Lecture
 - Next Class

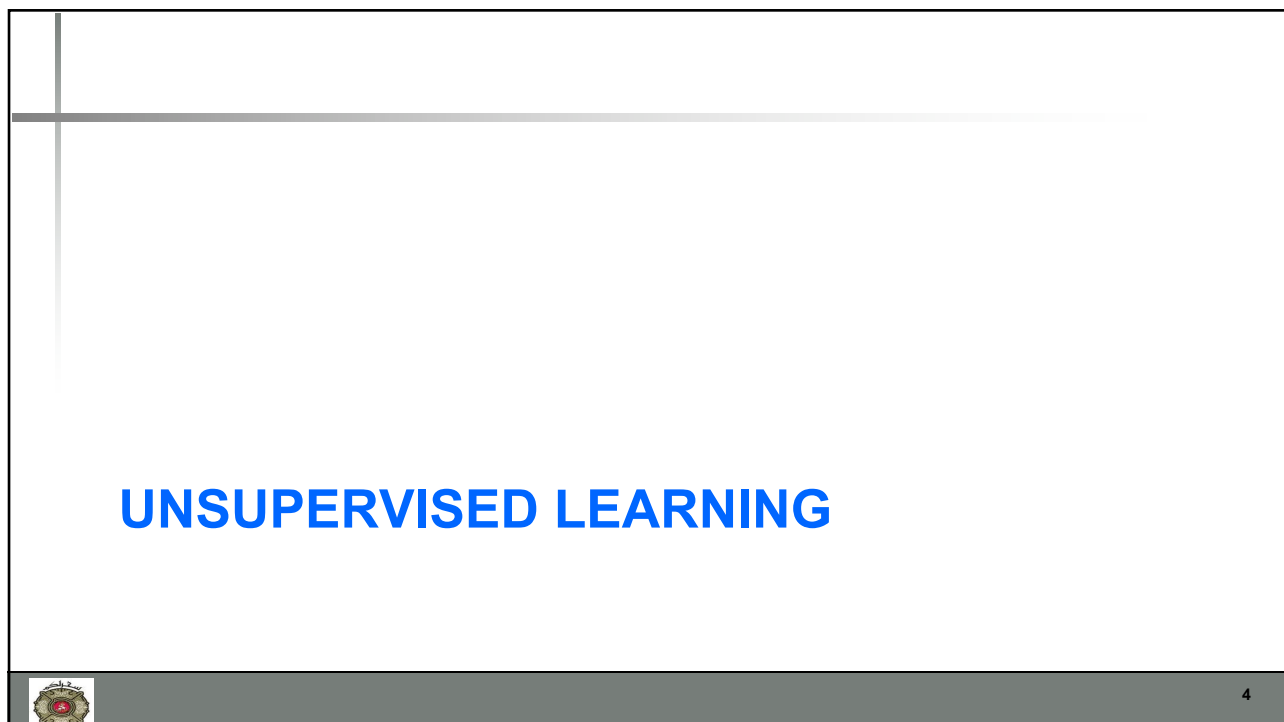
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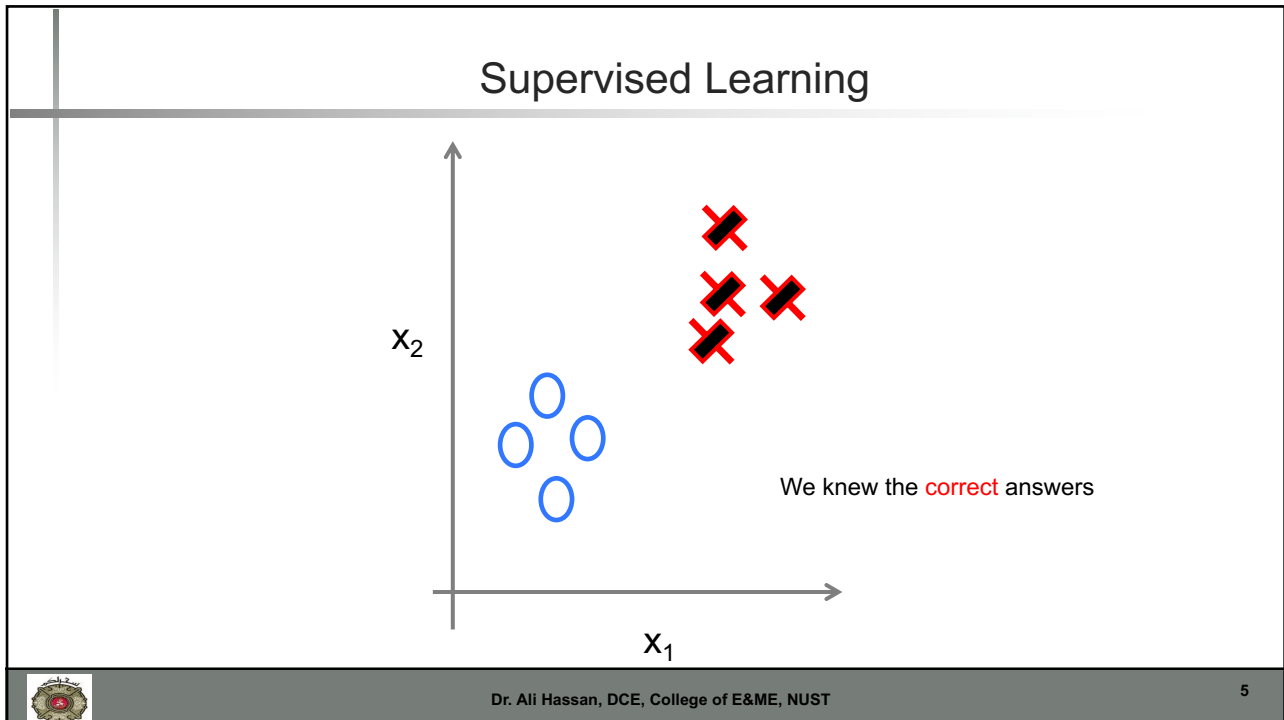
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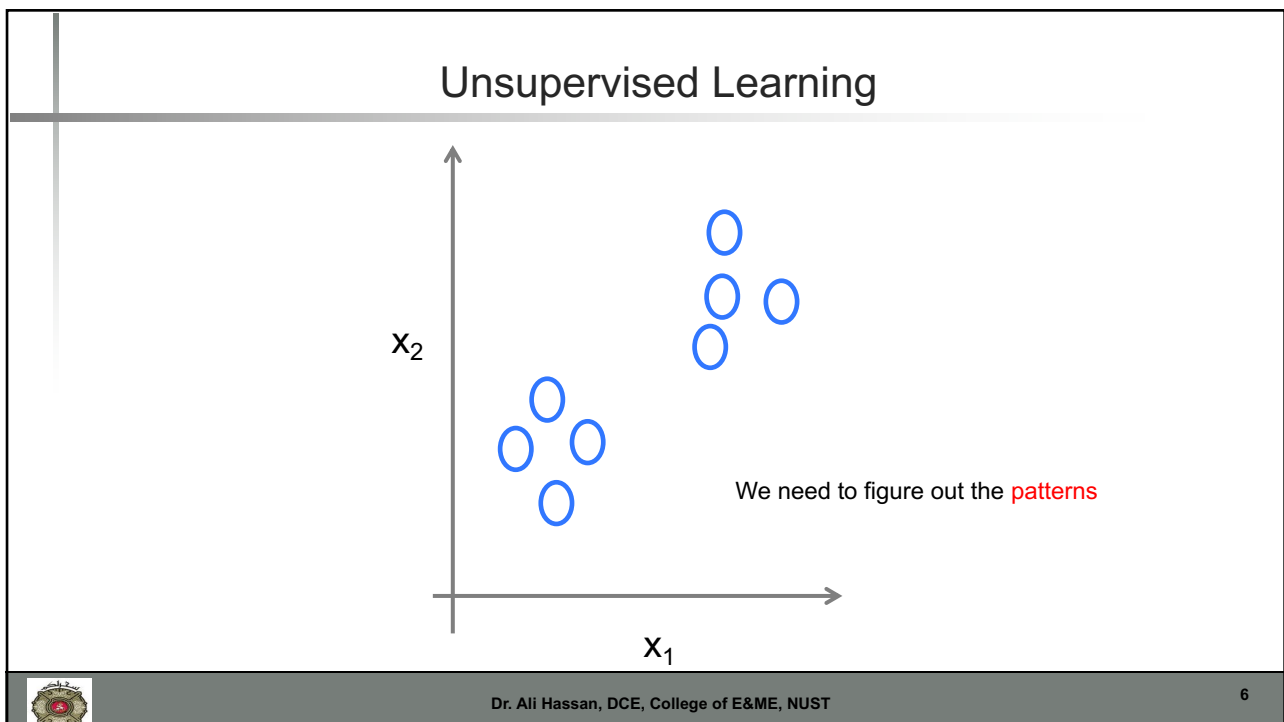
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Types of ML

- **Supervised Learning:**
 - You give to the computer some pairs of inputs/outputs, so in the future new when new inputs are presented you have an intelligent output.
- **Unsupervised Learning:**
 - You let the computer learn from the data itself without showing what is the expected output.

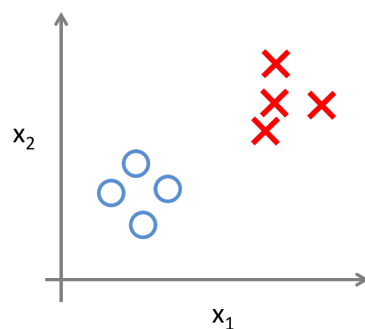
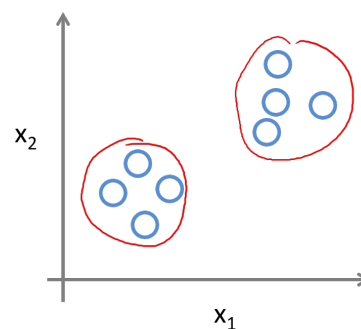


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7

7

Types of ML

Supervised Learning**Unsupervised Learning**

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8

8

An Application of Unsupervised Learning

Your briefing

Wednesday, 12 November

19°C
Good Weather

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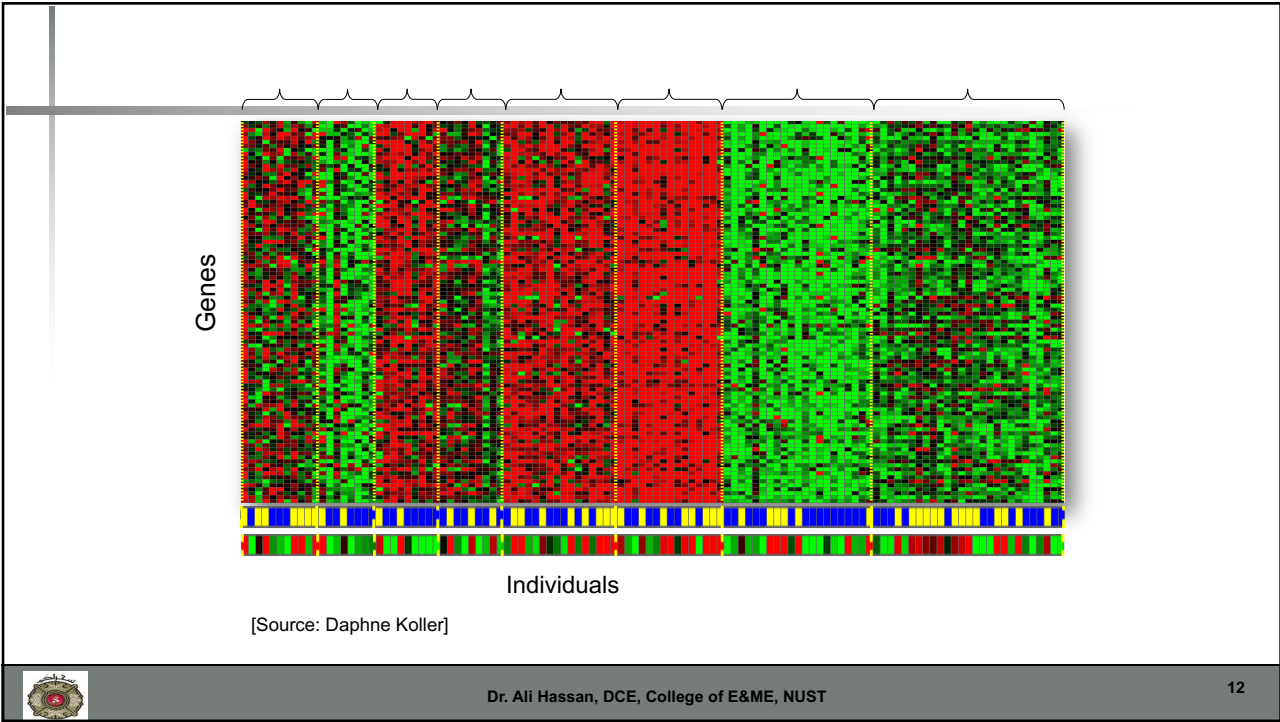
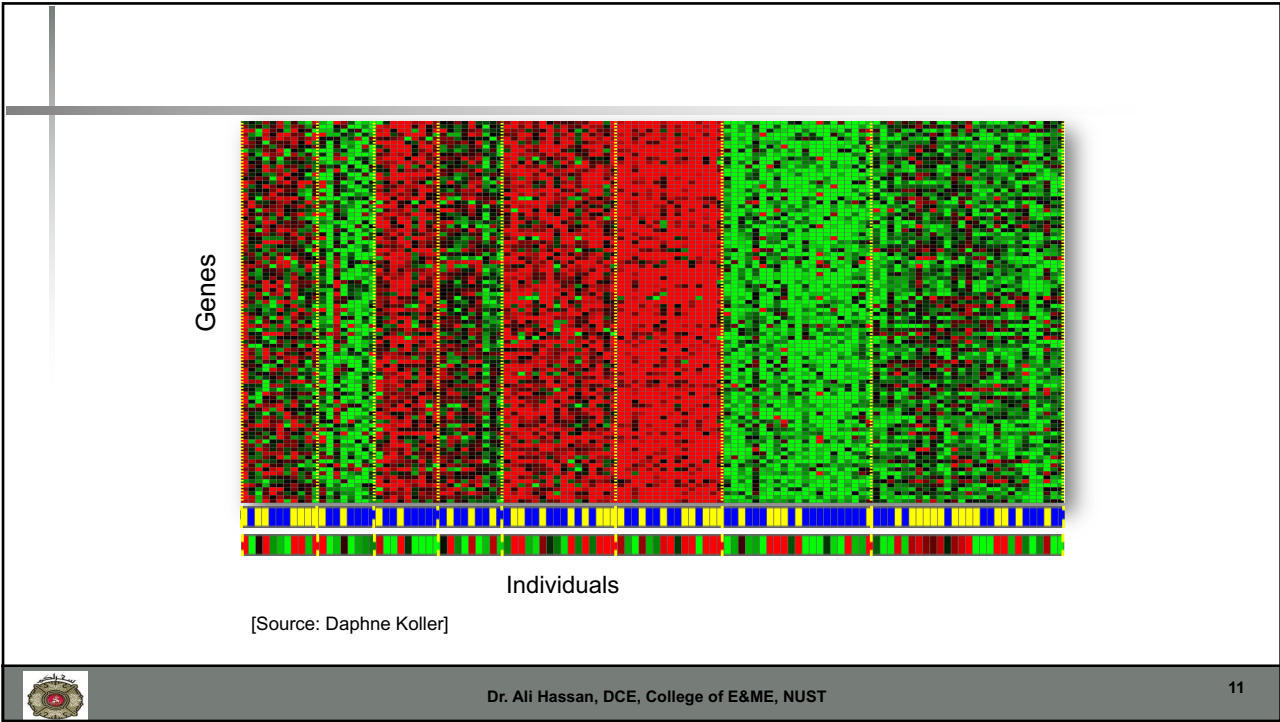
Context-based sentiment analysis using a BiGRU+BiLSTM-fusion...

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9

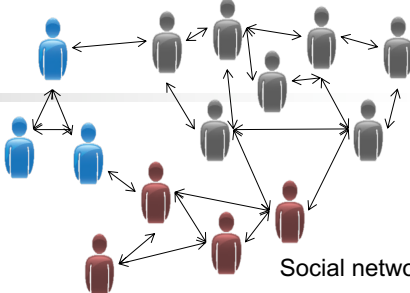
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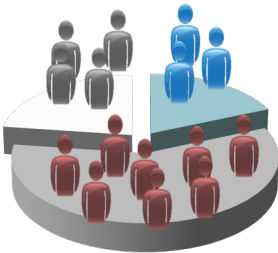




Organize computing clusters



Social network analysis



Market segmentation



Astronomical data analysis



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13

13

BBC Sign in Home News Sport Reel Worklife

NEWS



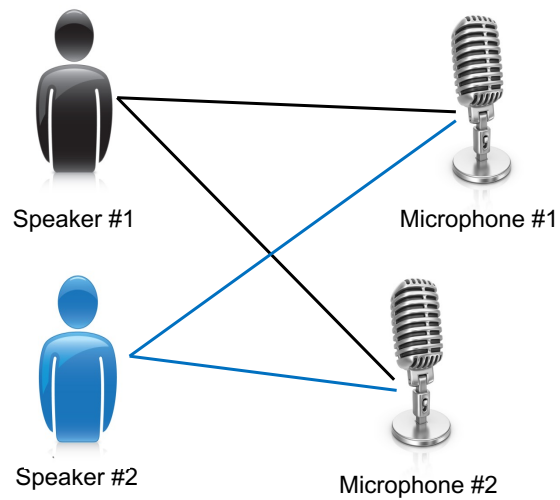


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14

14

Cocktail party problem



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15

15

Microphone #1:  Output #1: 
Microphone #2:  Output #2: 



[Audio clips courtesy of Te-Won Lee] Dr. Ali Hassan, DCE, College of E&ME, NUST

16

16

Cocktail party problem algorithm

```
[W,s,v] = svd((repmat(sum(x.*x,1),size(x,1),1).*x)*x');
```

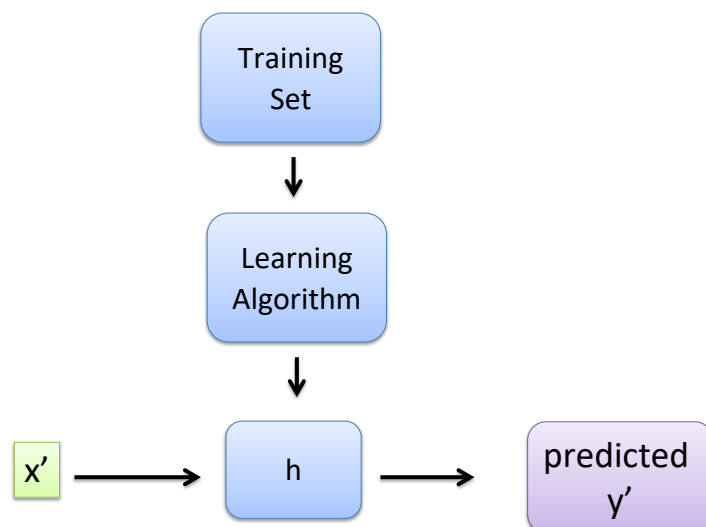


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17

17

Supervised Learning



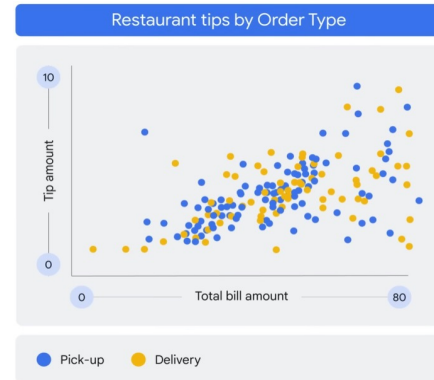
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18

18

Example: Supervised Learning

- Supervised learning works on already labelled data
 - Learn from past examples to classify current values

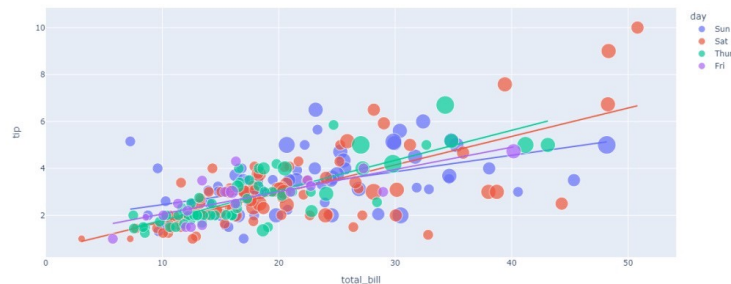


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19

19

Example: Supervised Learning



- total_bill: Total bill in dollars including taxes
- tip: Tip given to waiters in dollars
- sex: gender of the person paying the bill
- smoker: whether the person smoked or not
- day: day of the week
- time: lunch or dinner
- size: number of people in a table

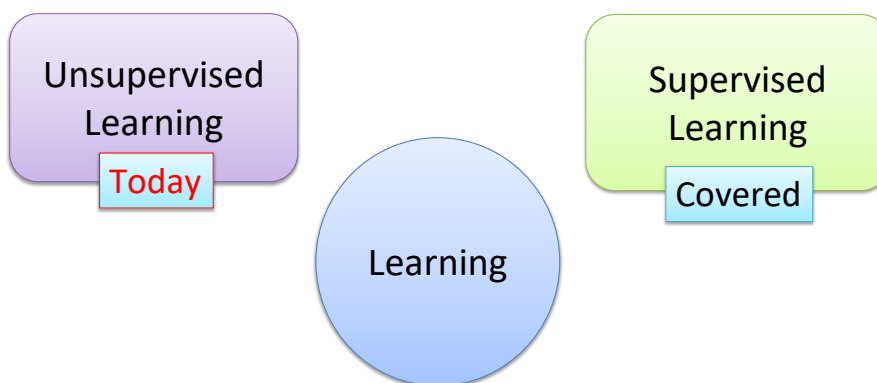


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20

20

Pattern Recognition

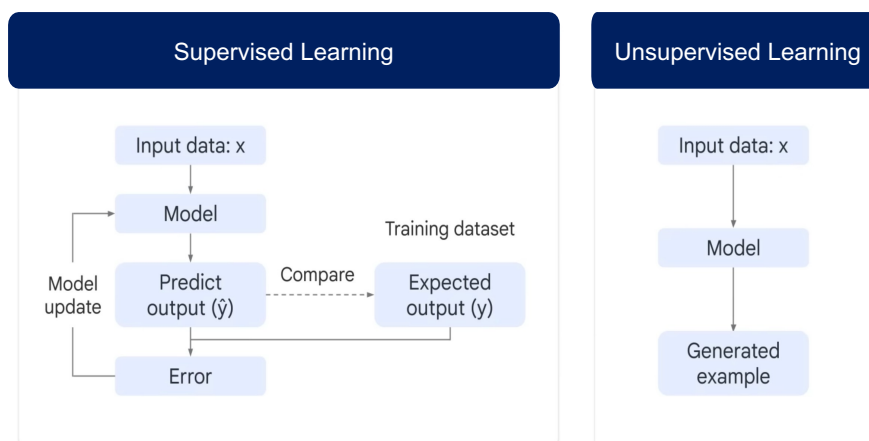


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21

21

Supervised vs Unsupervised



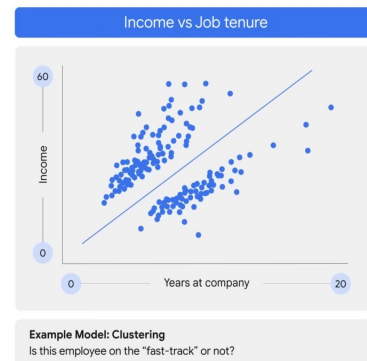
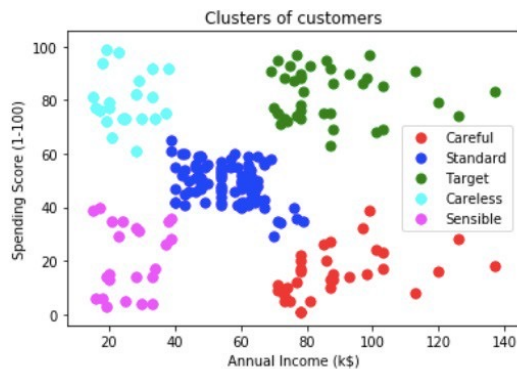
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22

Example: Unservised Learning

- Unsupervised learning works on data that is not labelled
 - Finds natural patterns in raw data

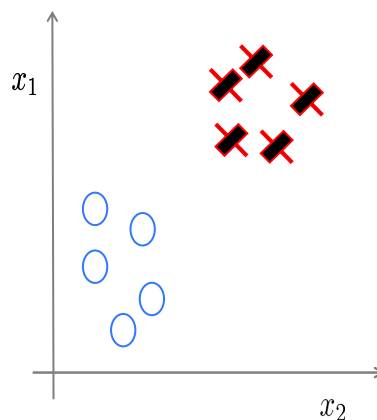


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23

23

Supervised learning



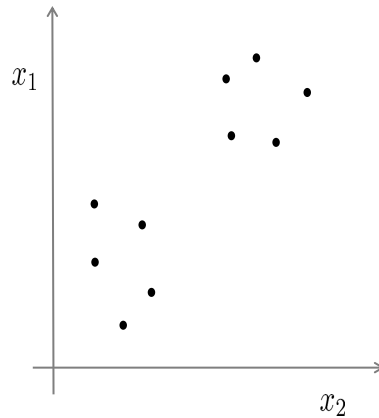
Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, (x^{(m)}, y^{(m)})\}$



24

24

Unsupervised learning



Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(m)}\}$



25

25

Motivation

- There are patterns (hidden) in nature
- Unsupervised learning extracts these patterns

Main Idea

- Similar data is collocated in the dimensional space
- Data points within a group are more similar than those outside the group



26

26

The Great American Devide



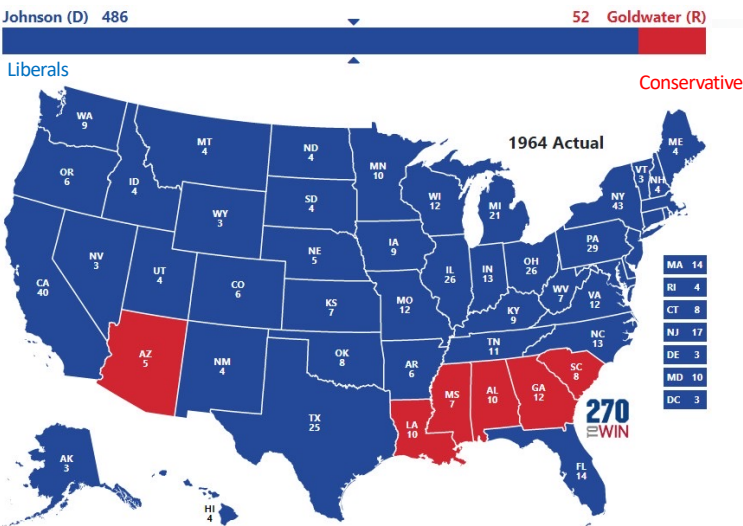
- More than two-thirds of **Republicans** identify as conservative (77%).
- Just over half of **Democrats** identify as liberal (58%).



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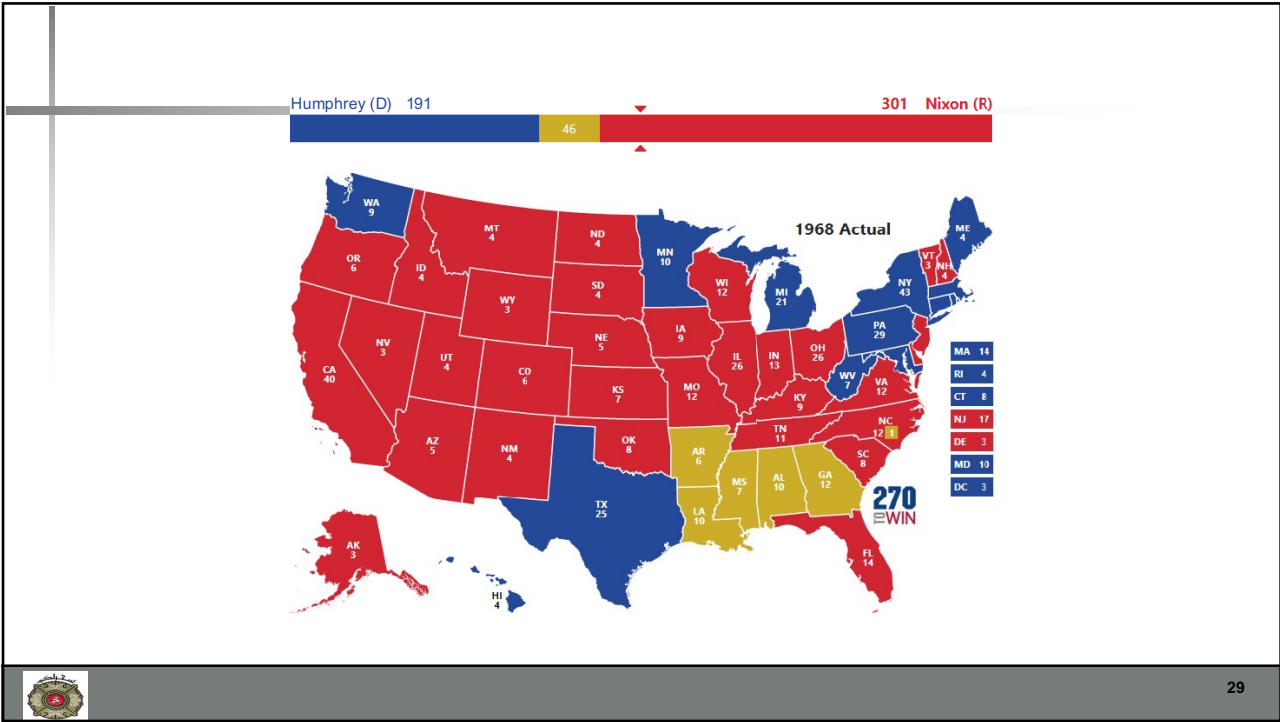
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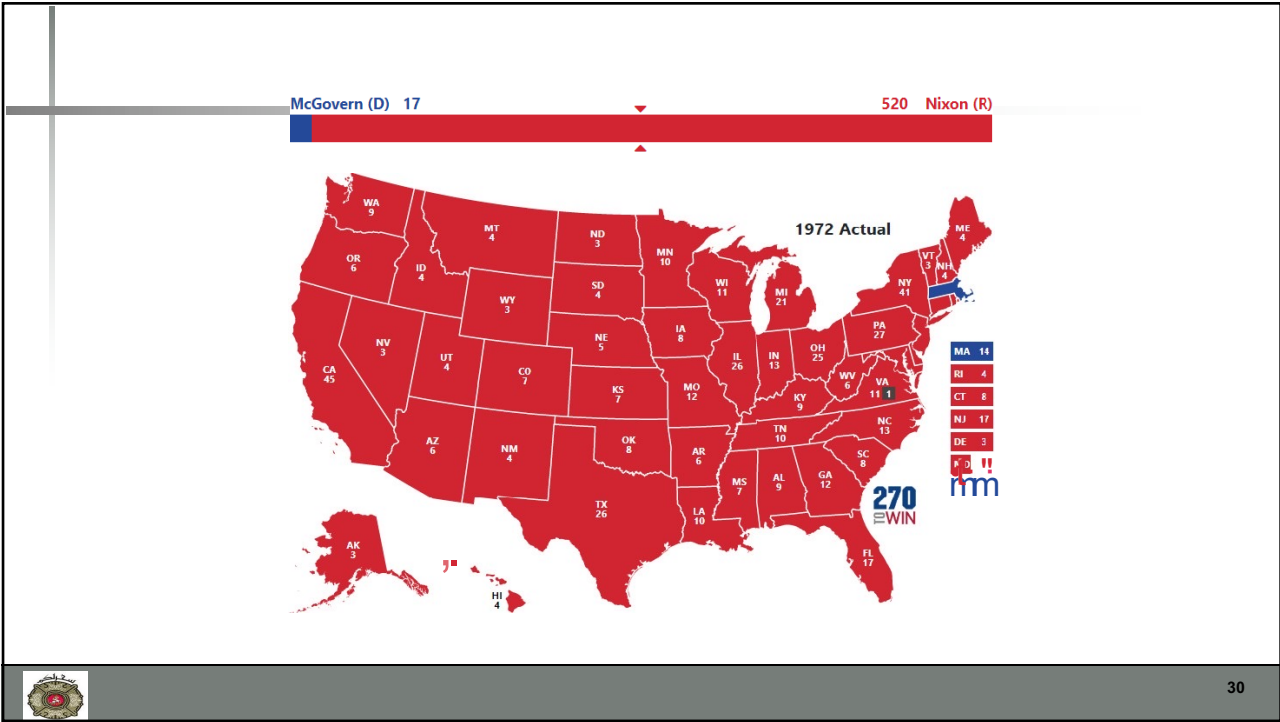


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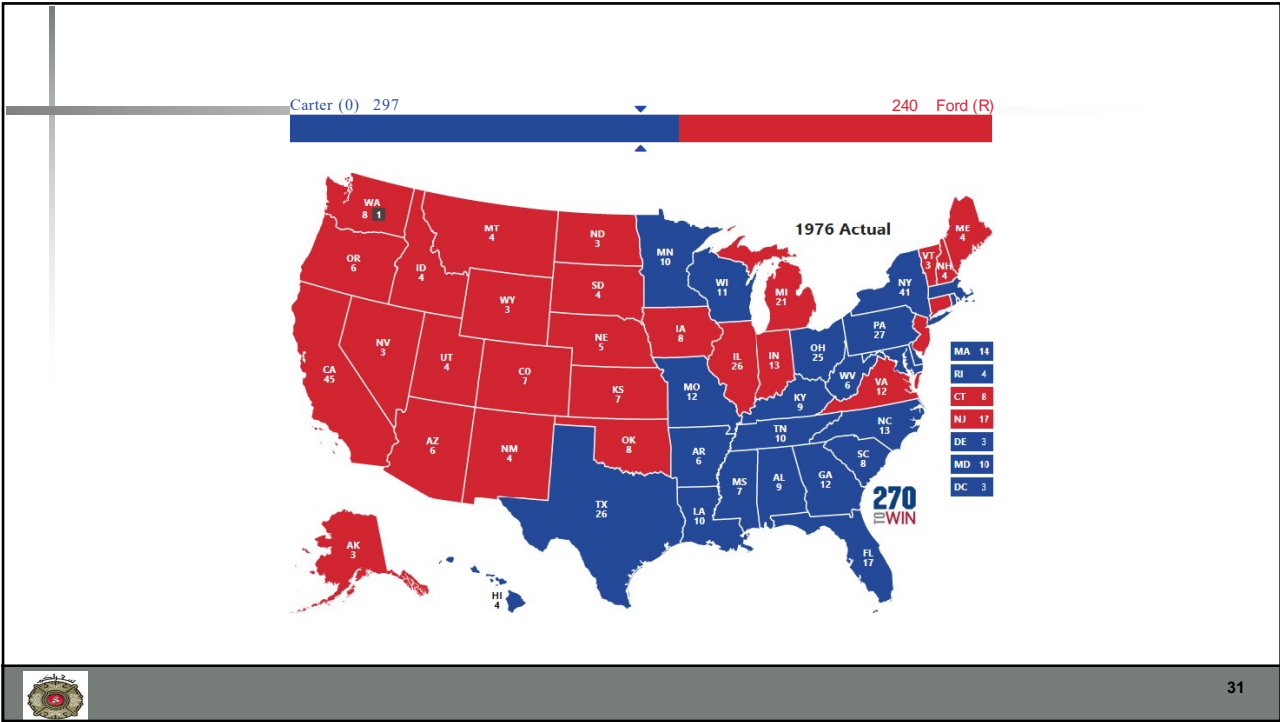
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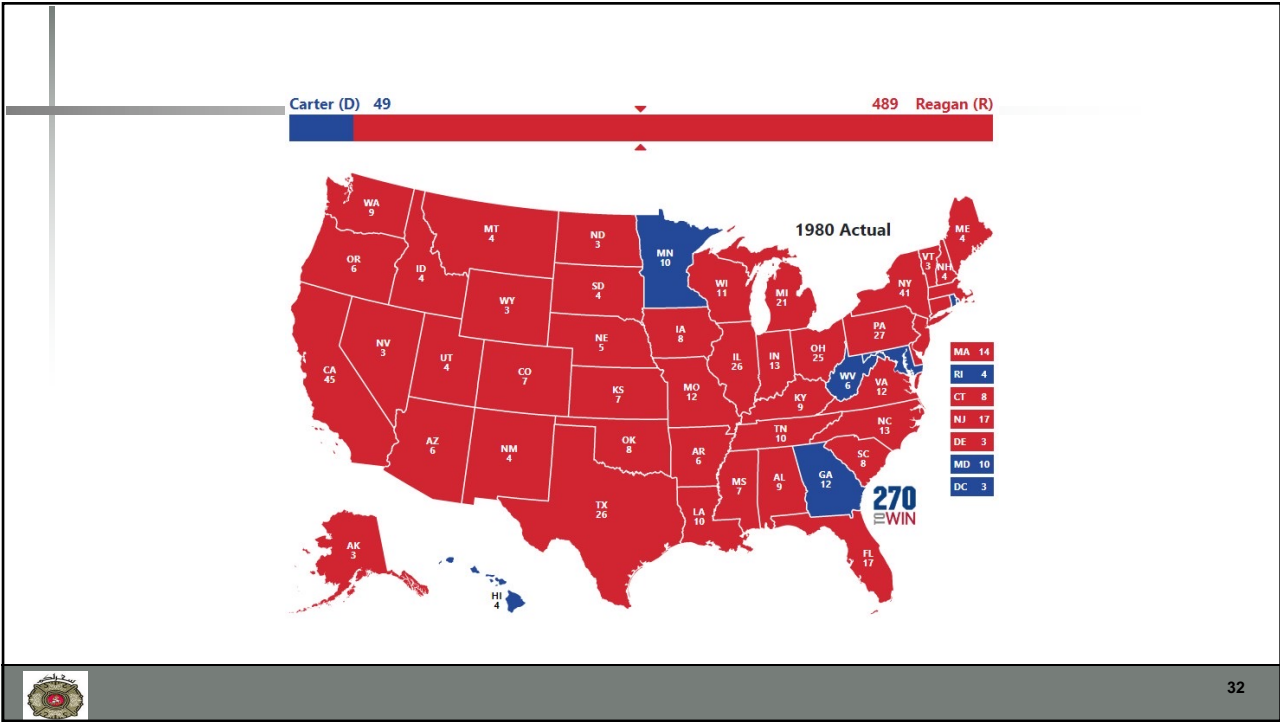
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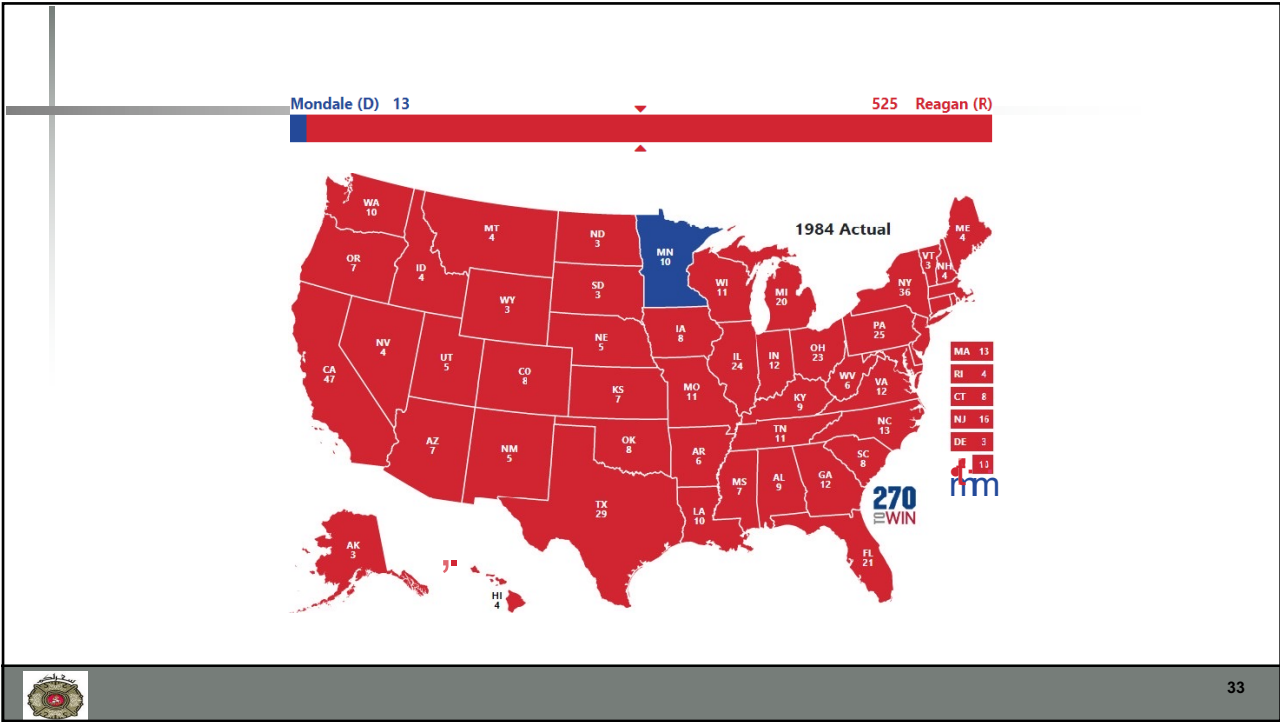
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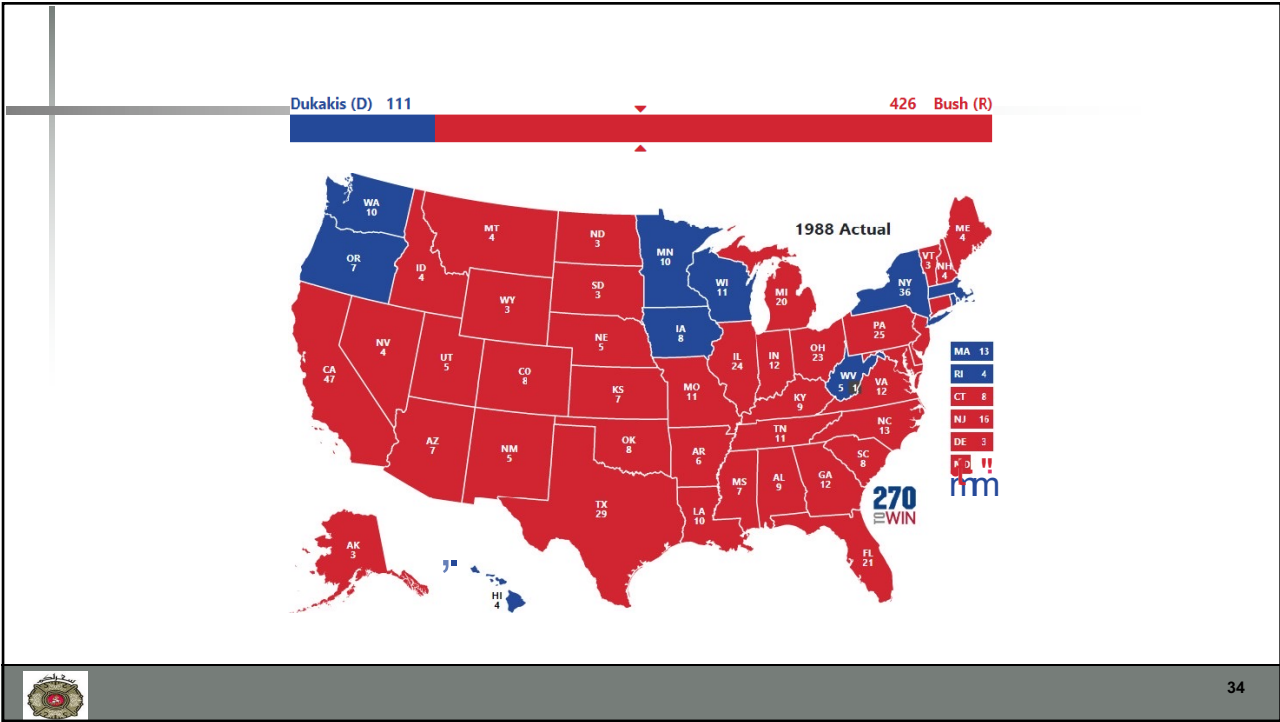
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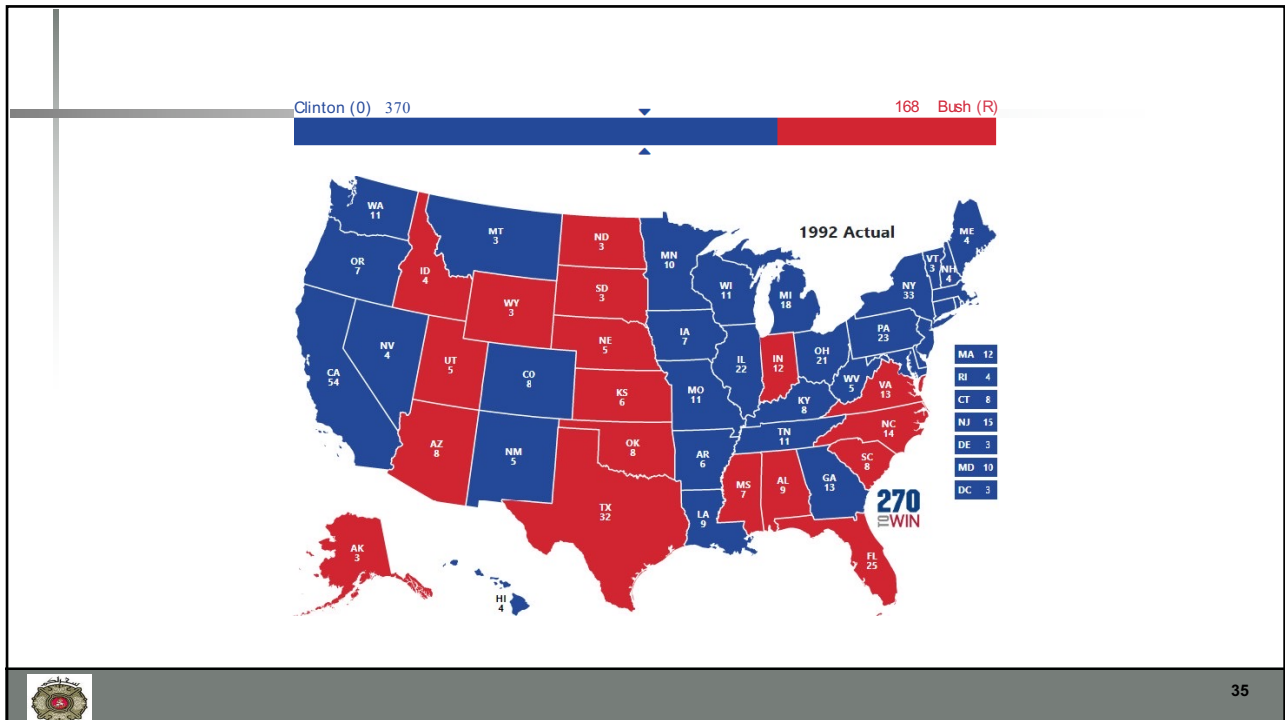
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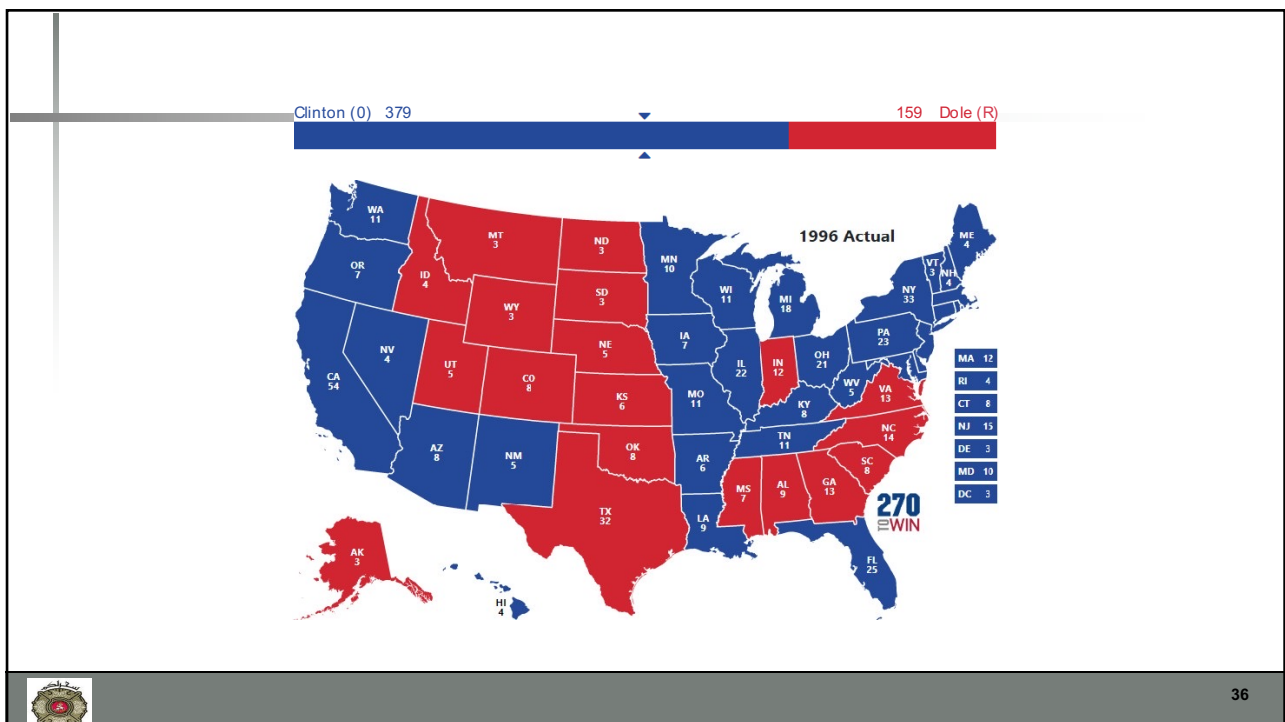
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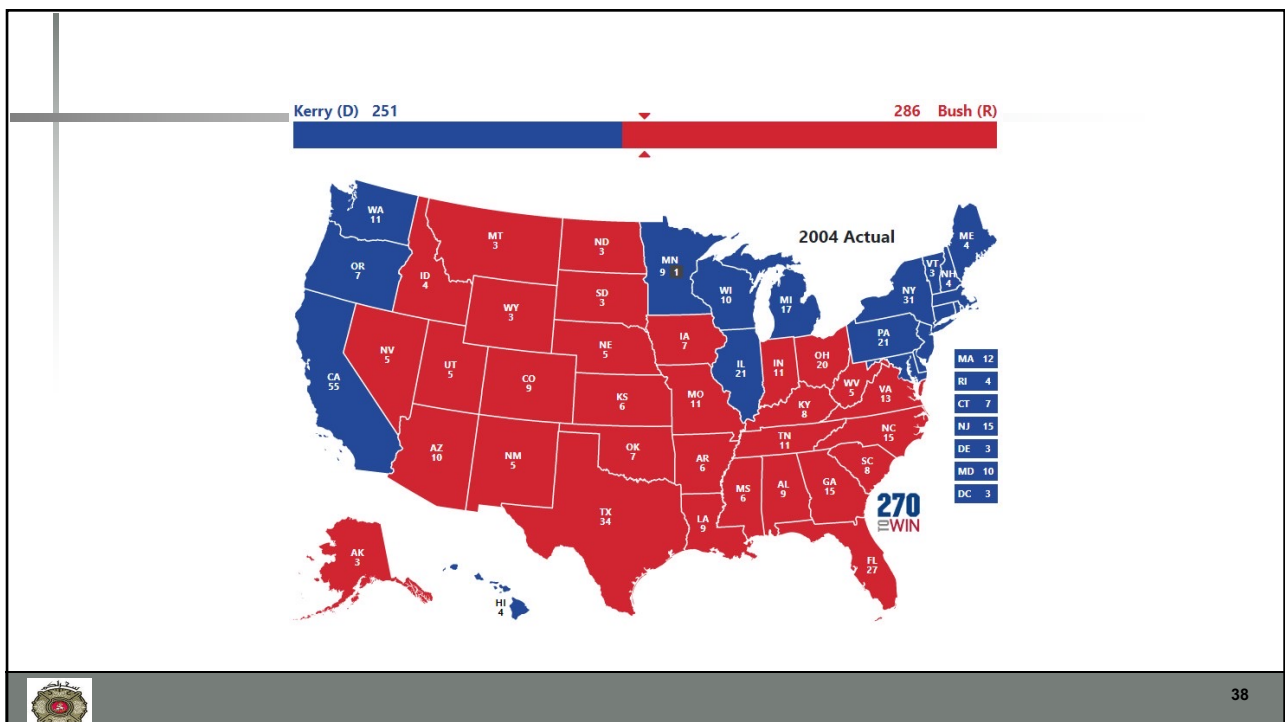
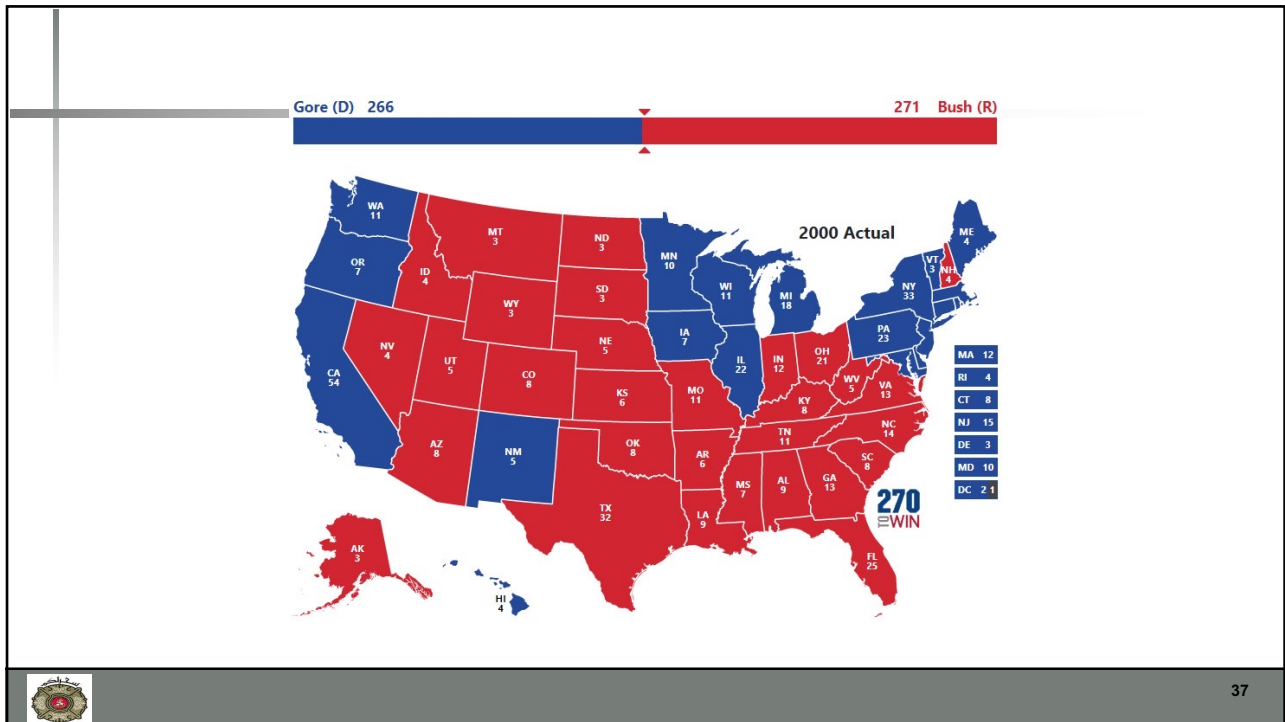
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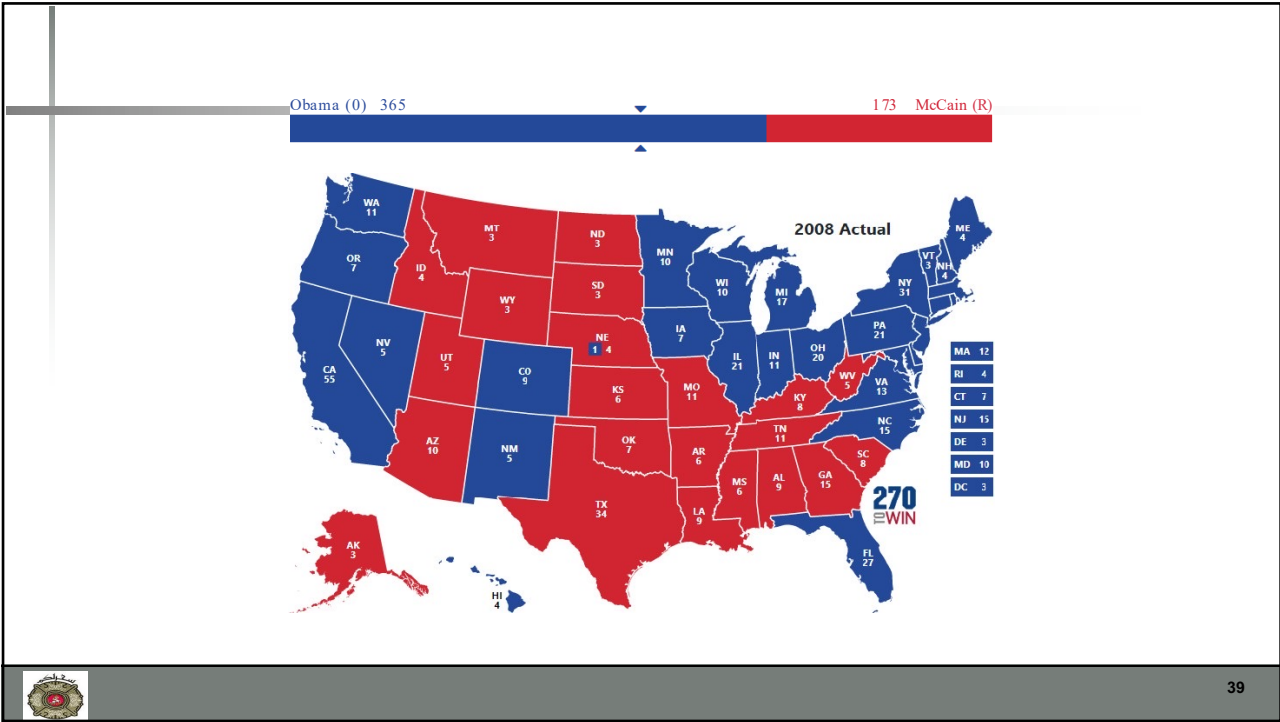


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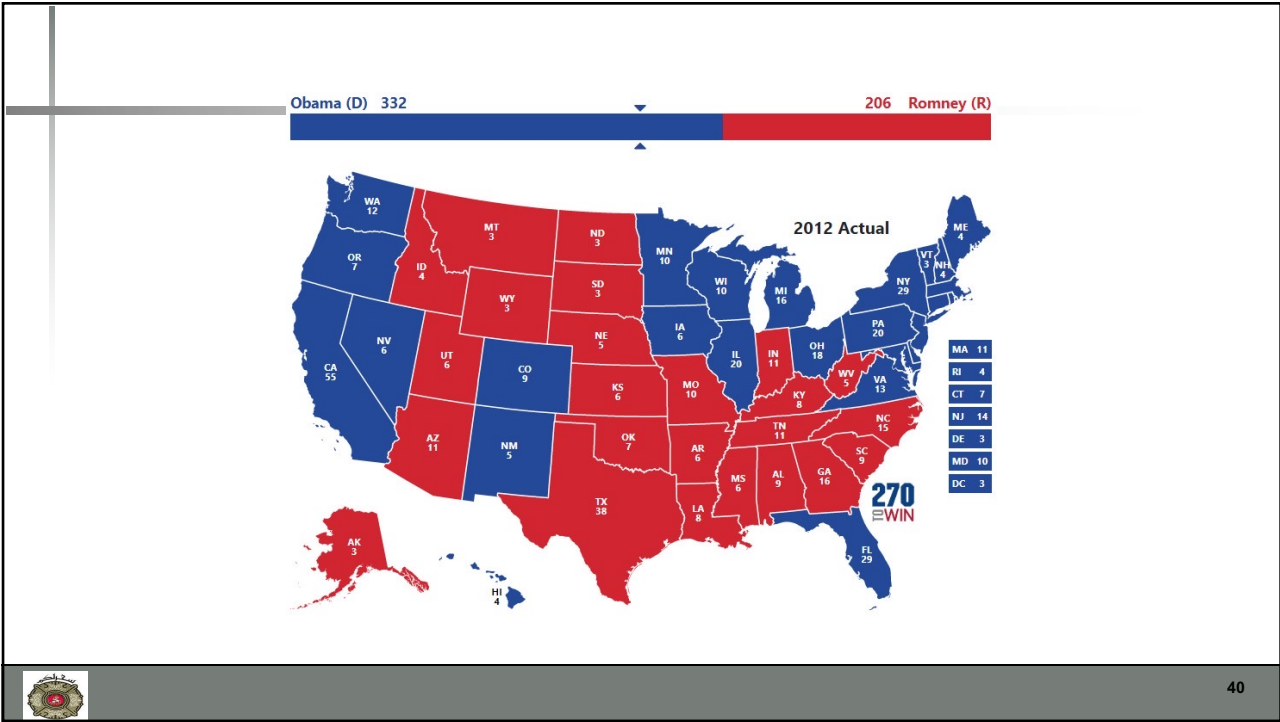


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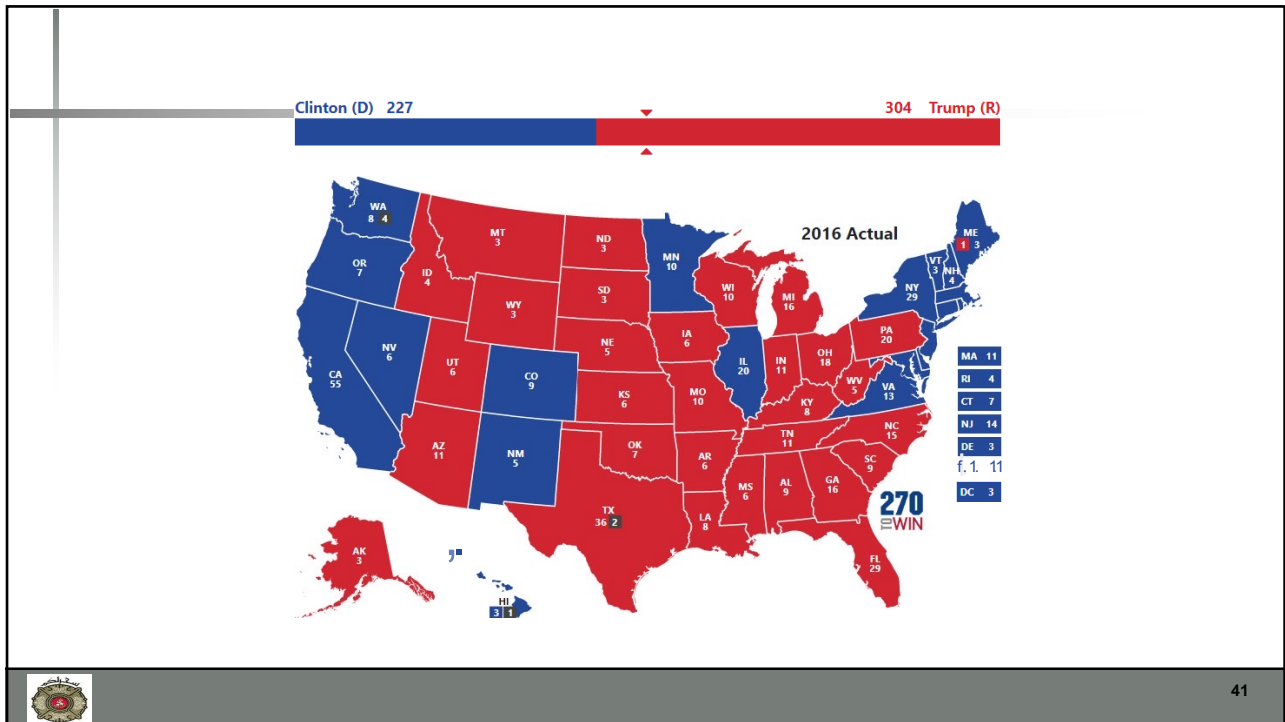




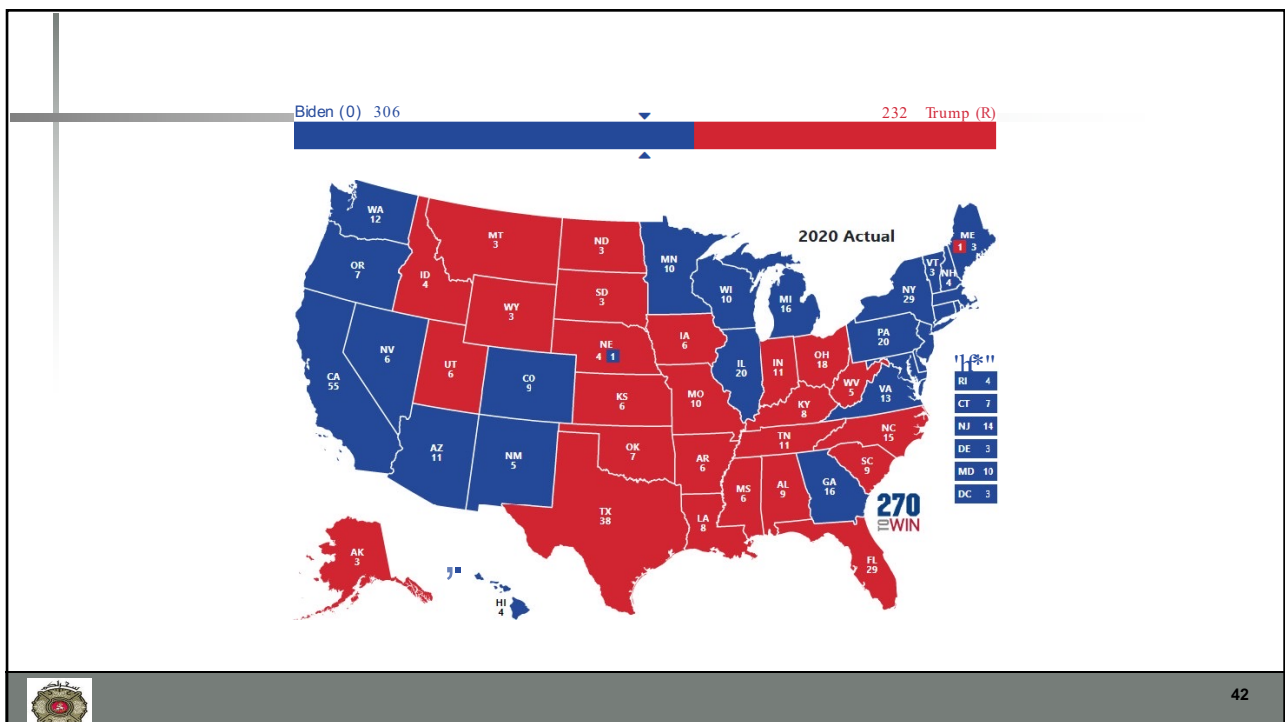
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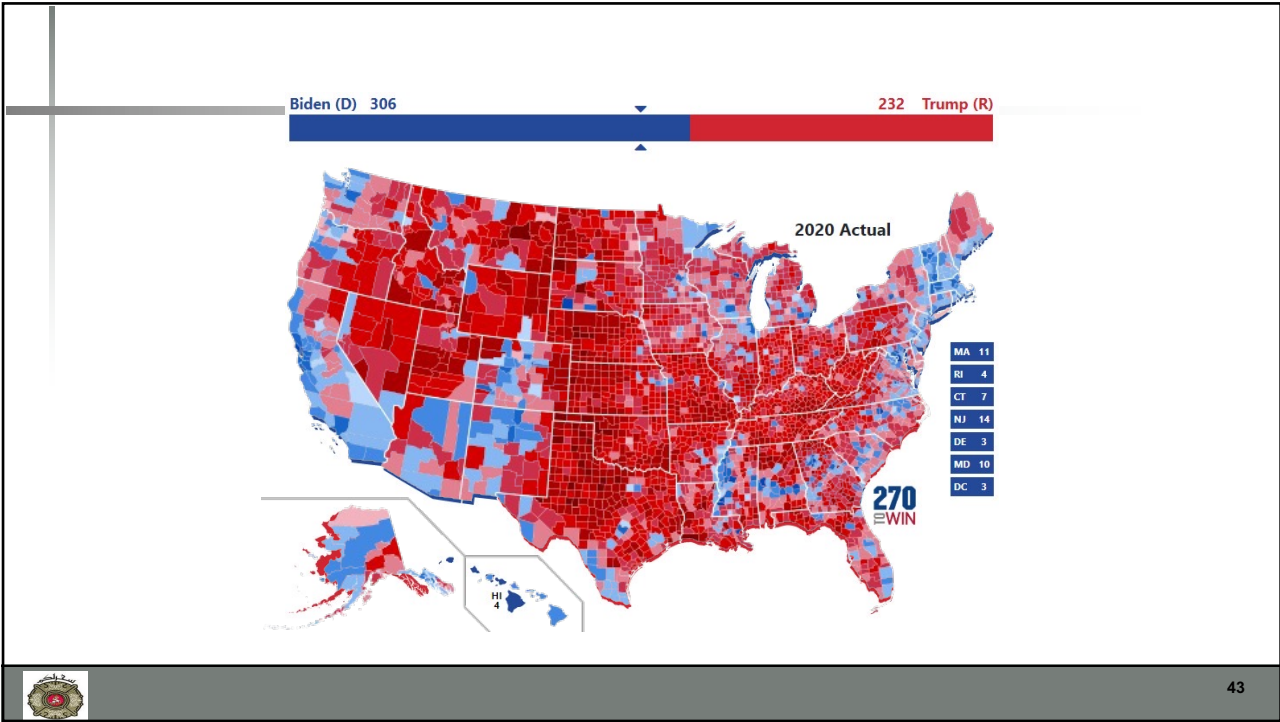
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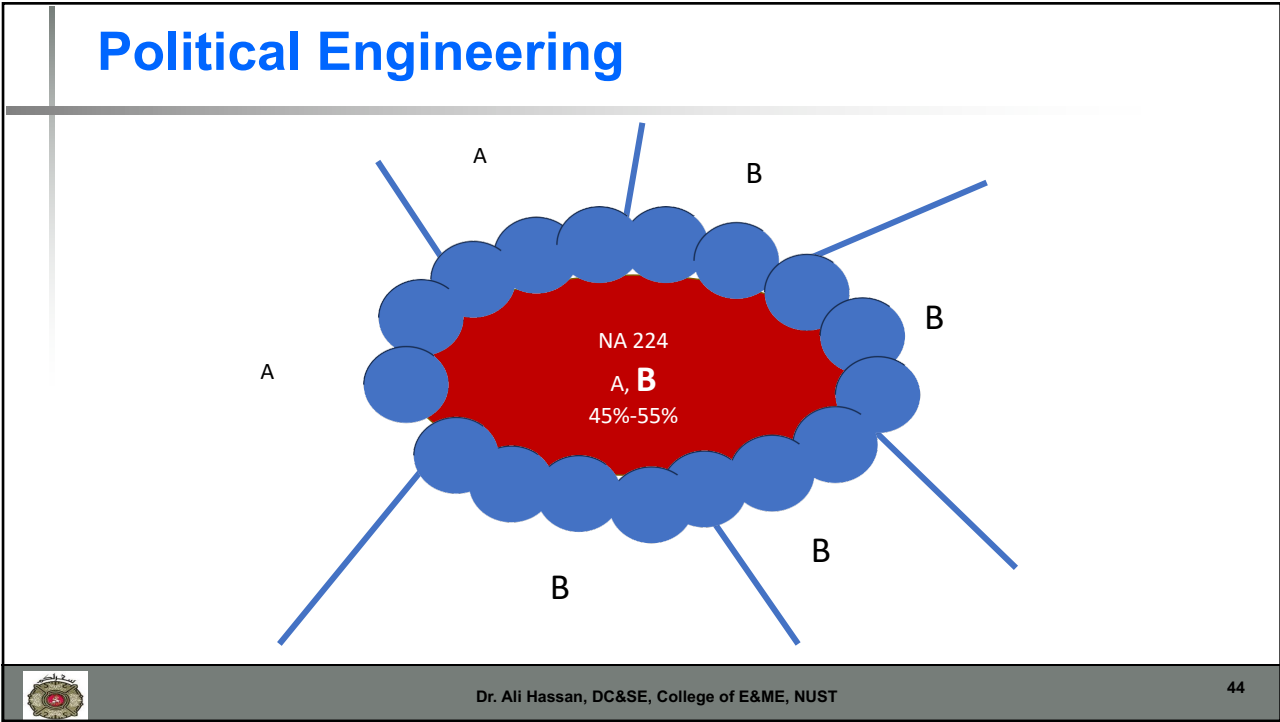
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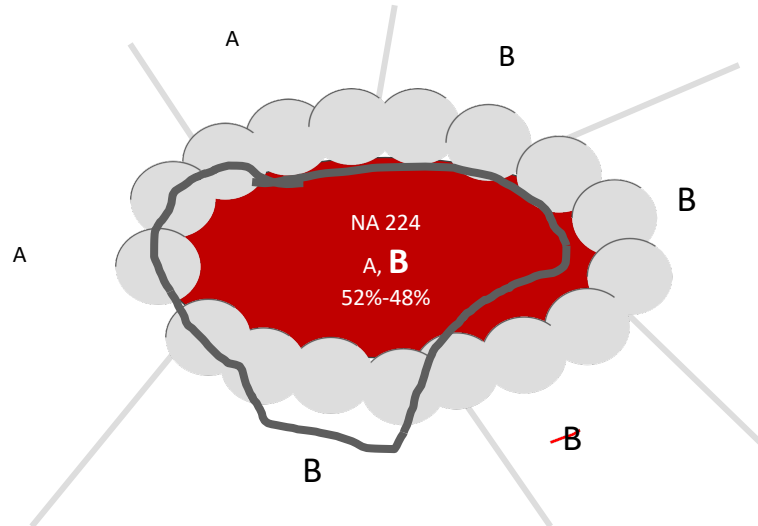


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44

New Halqa Bandiyan using Clustering



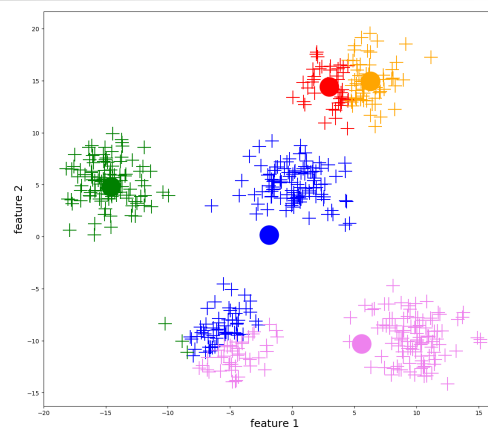
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45

45

Clustering

- split objects into clusters
- describe clusters by centroids
- need similarity (distance)
- most widely used: k-means



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65

65

K-MEANS CLUSTERING

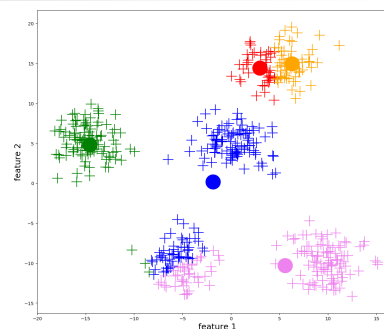


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k-means Clustering

- Unlabelled data
- Goal: split data into k disjoint groups
- Points in each cluster are closer to its center than to other centers
- Computes allocation by iteration



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67

67

K-means algorithm

- Input:
 - K (number of clusters)
 - Training set $\{x^{(1)}, x^{(2)}, \dots, x^{(m)}\}$
- $x^{(i)} \in \mathbb{R}^n$



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68

68

K-means algorithm

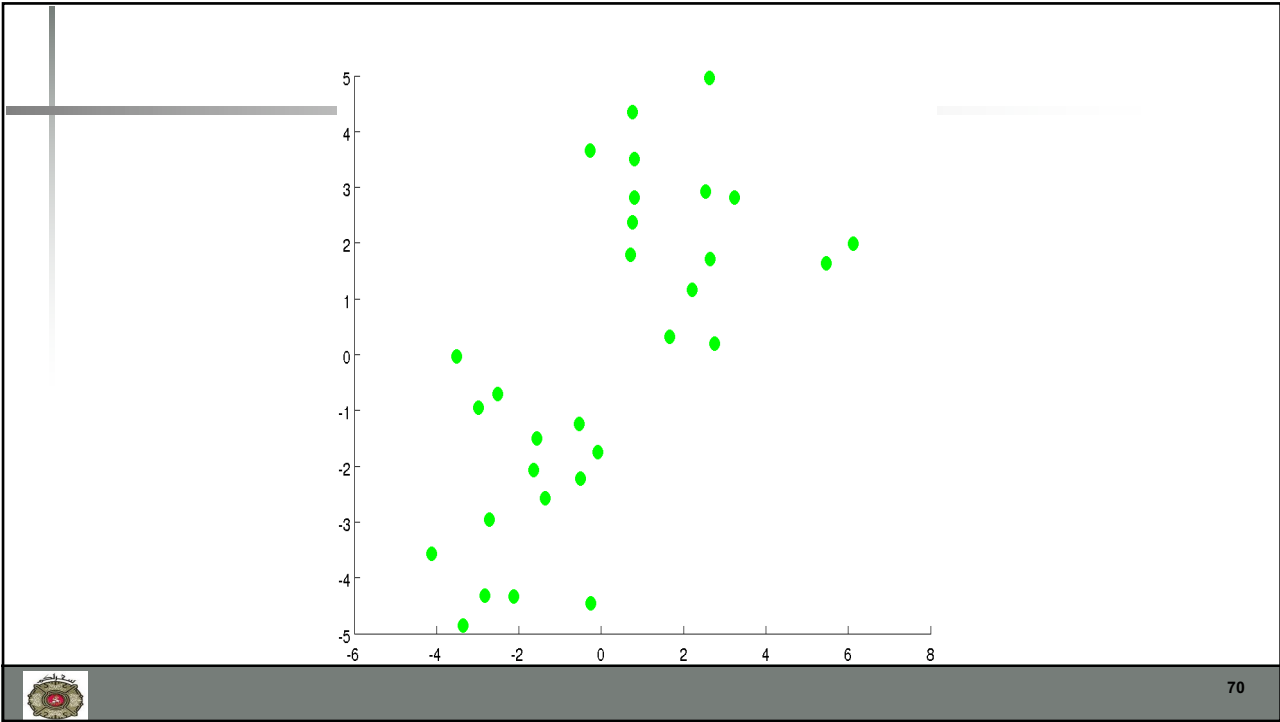
- Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K \in \mathbb{R}^n$
- Repeat {
 - Cluster Assignment {
 - for $i=1$ to m
 - $c^{(i)} :=$ index (1 to K) of cluster centroid closest to $x^{(i)}$
 - Centroid Movement {
 - for $k=1$ to K
 - $\mu_k :=$ mean of points assigned to cluster k



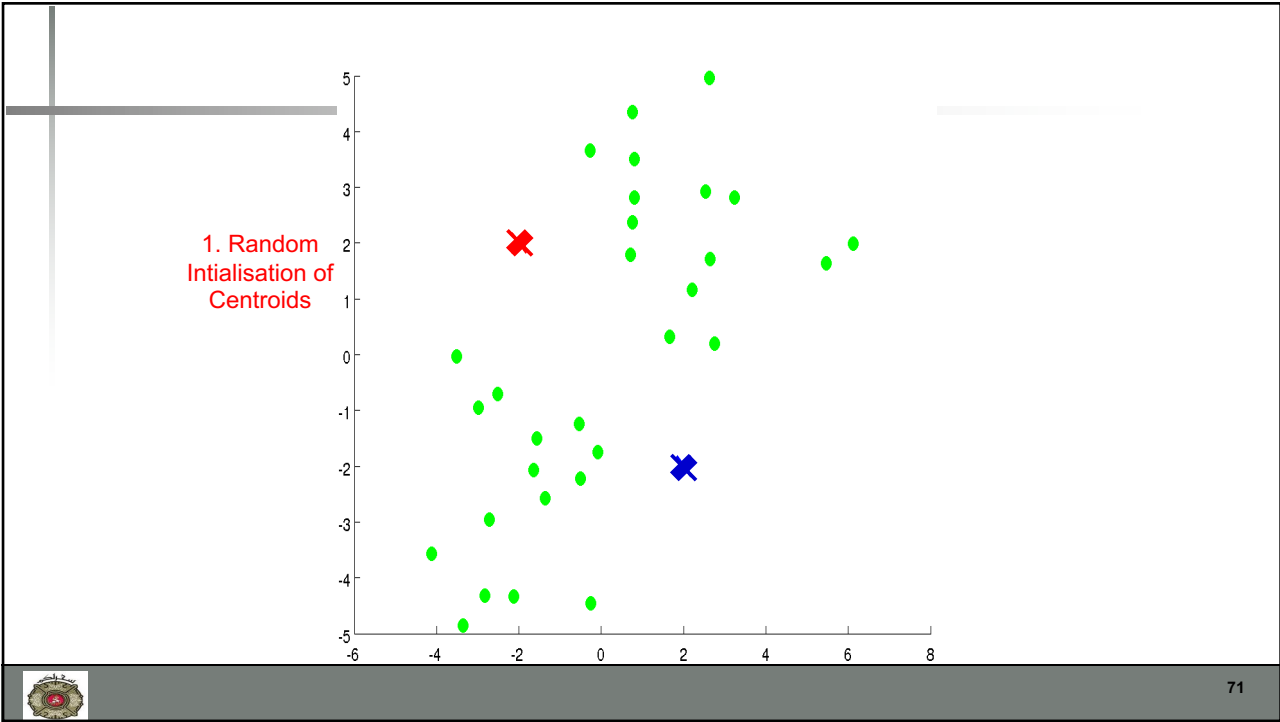
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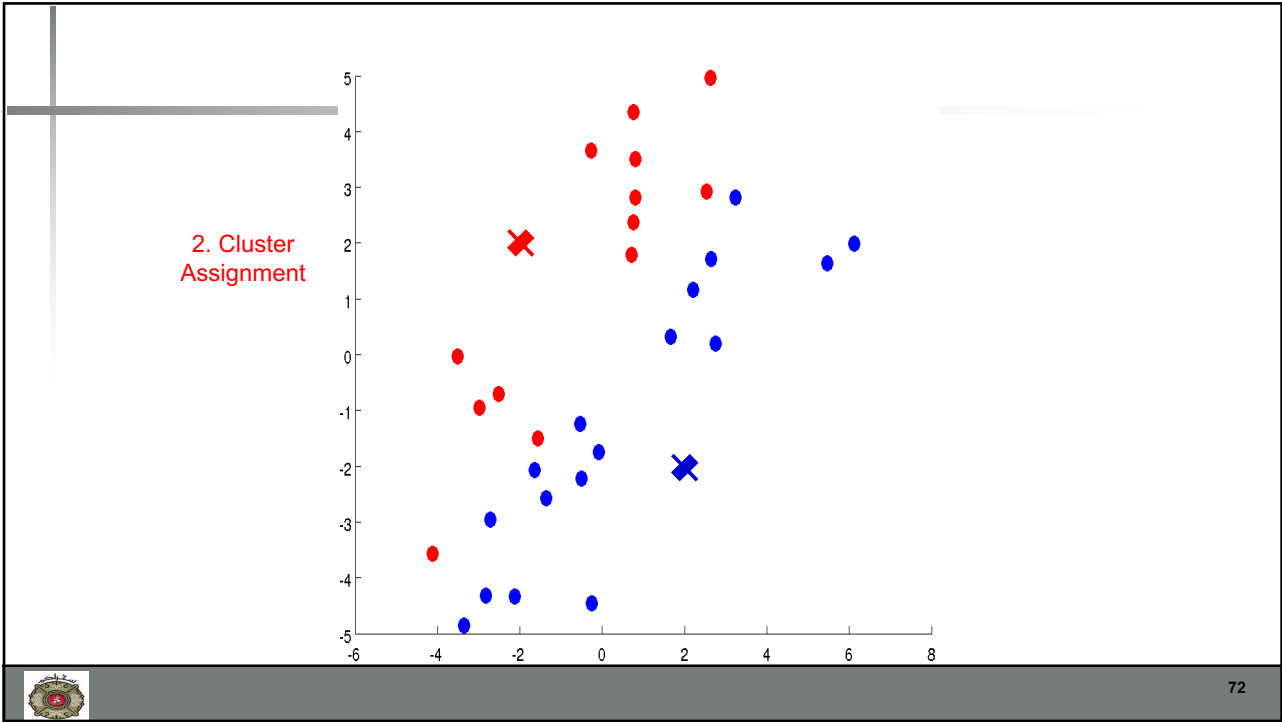
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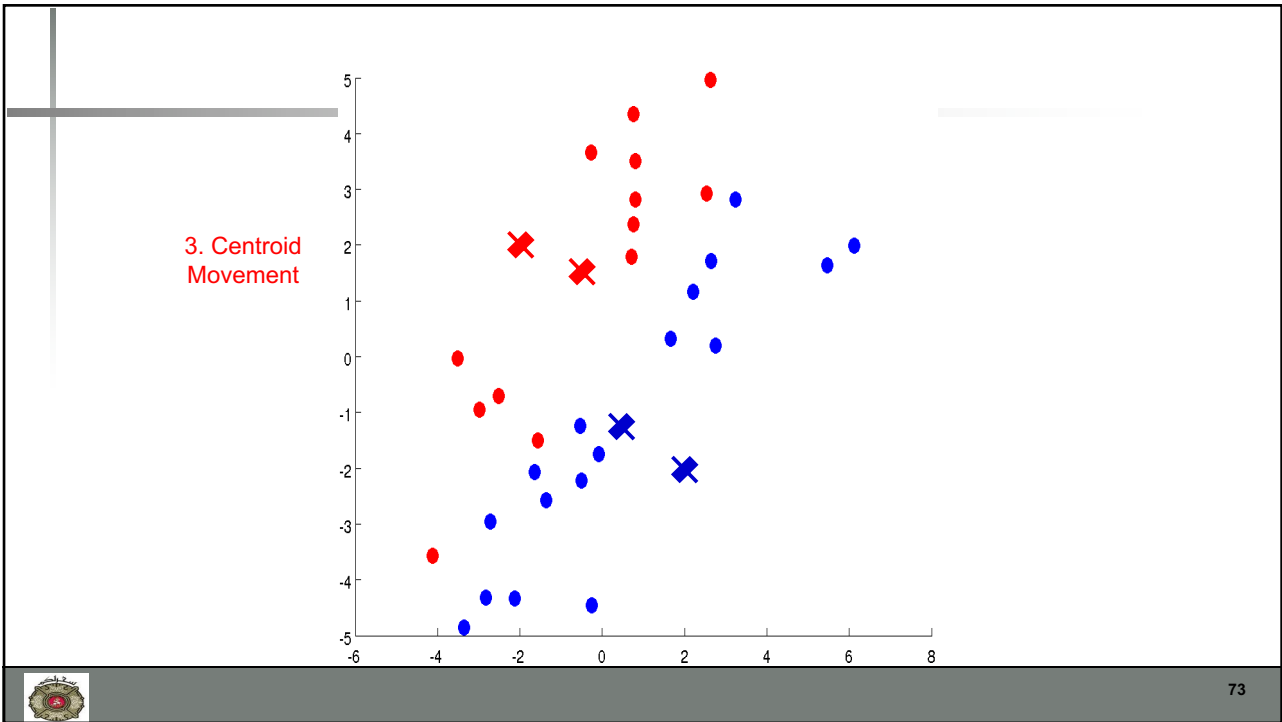
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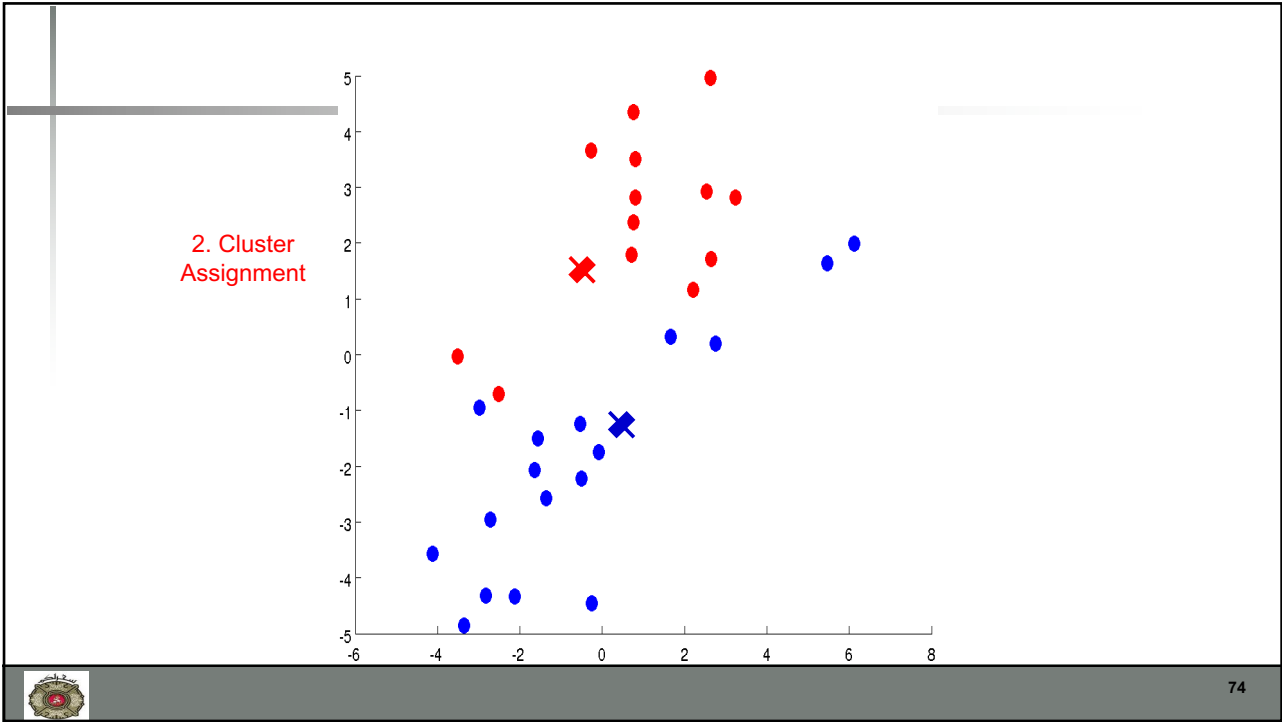
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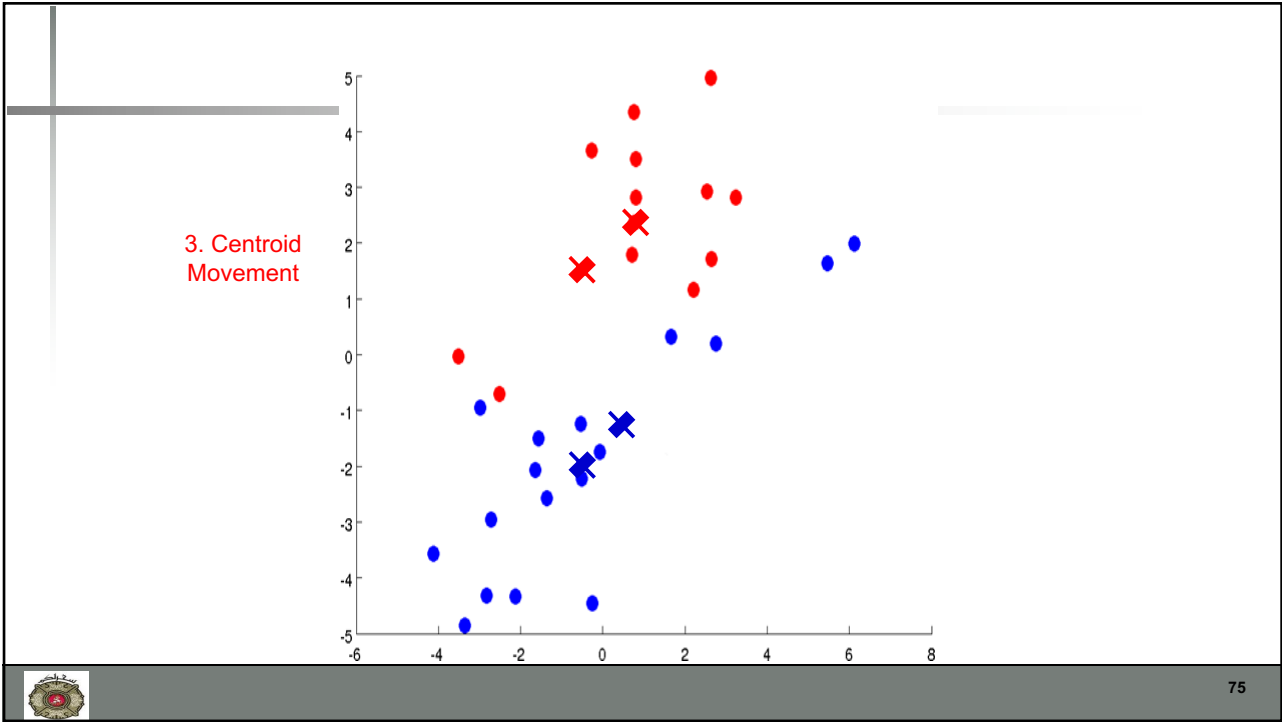
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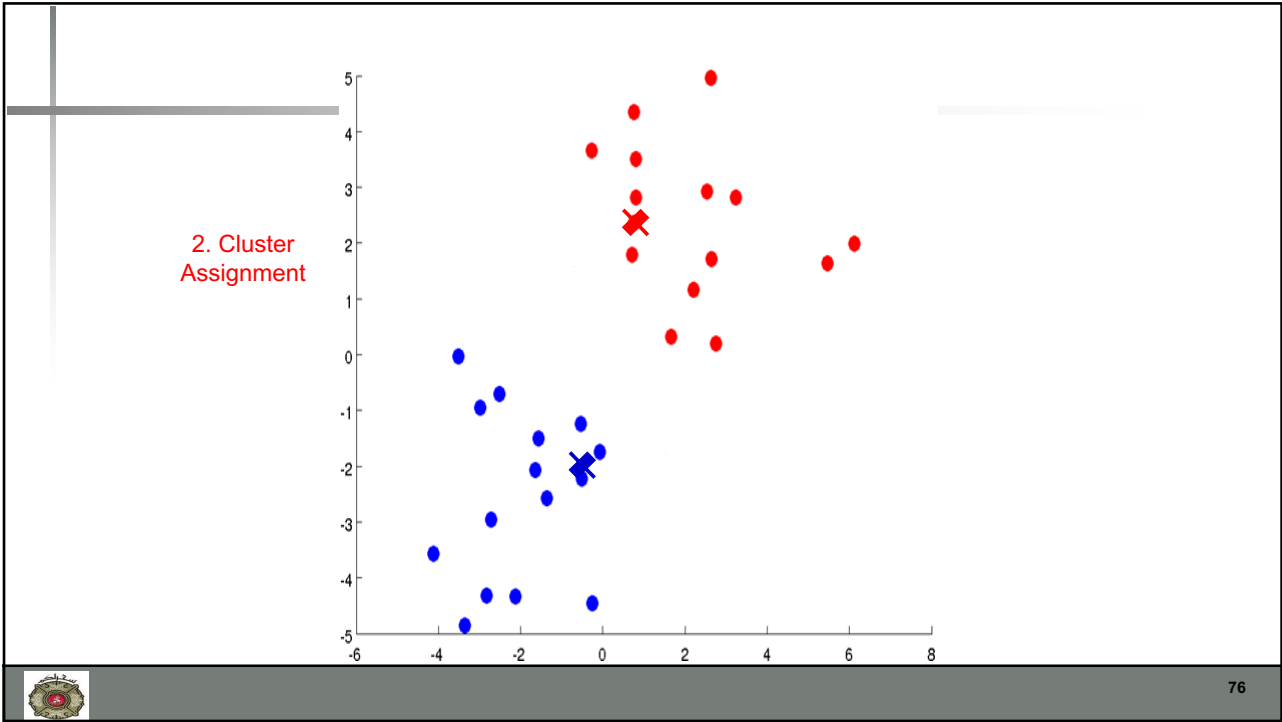
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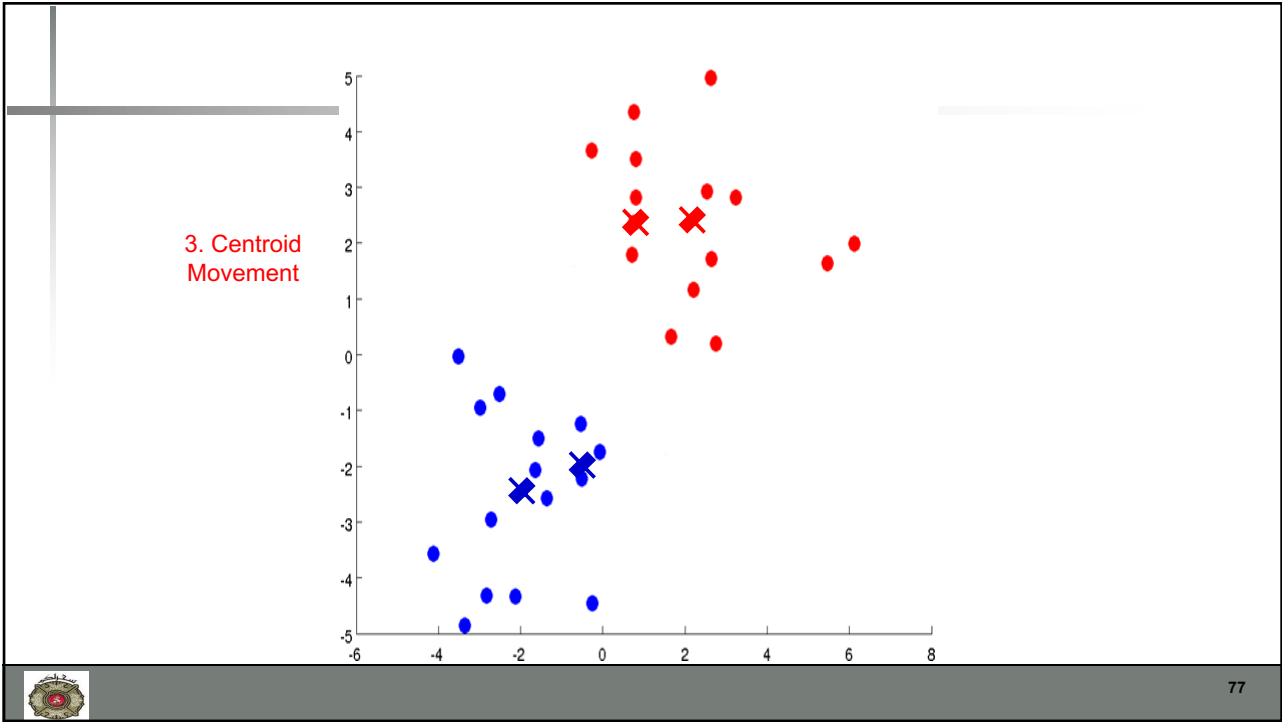
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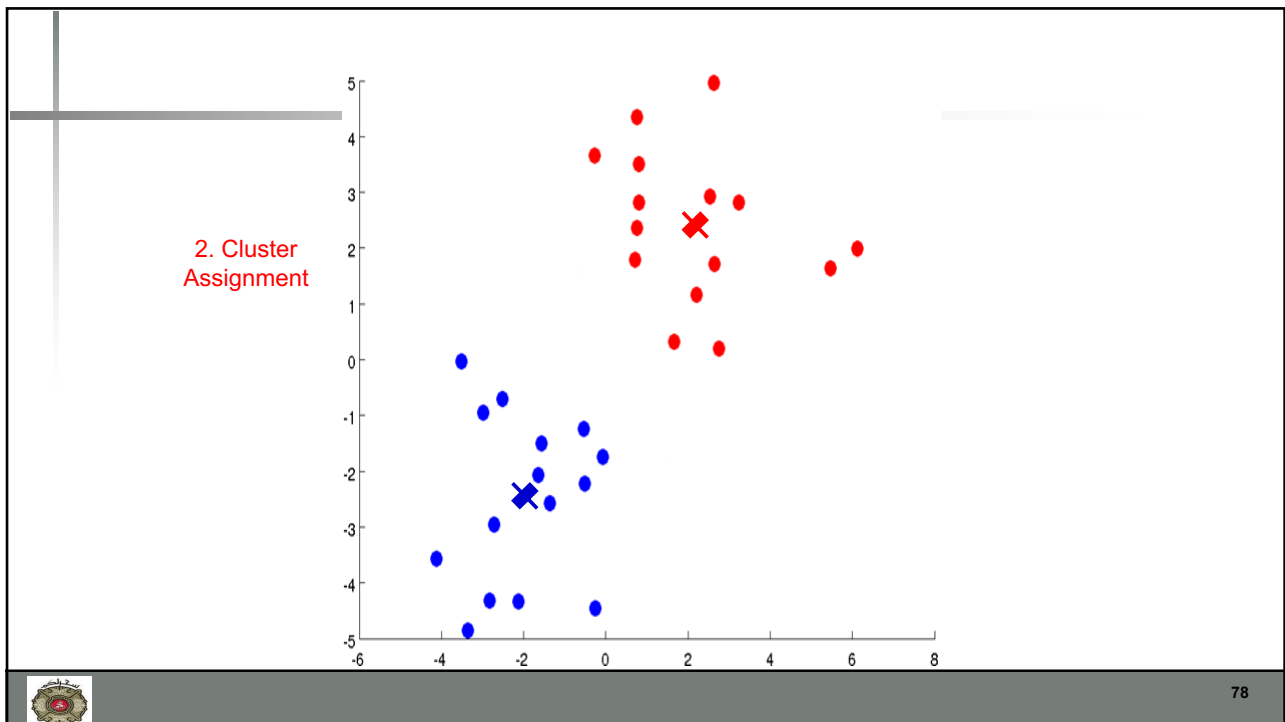
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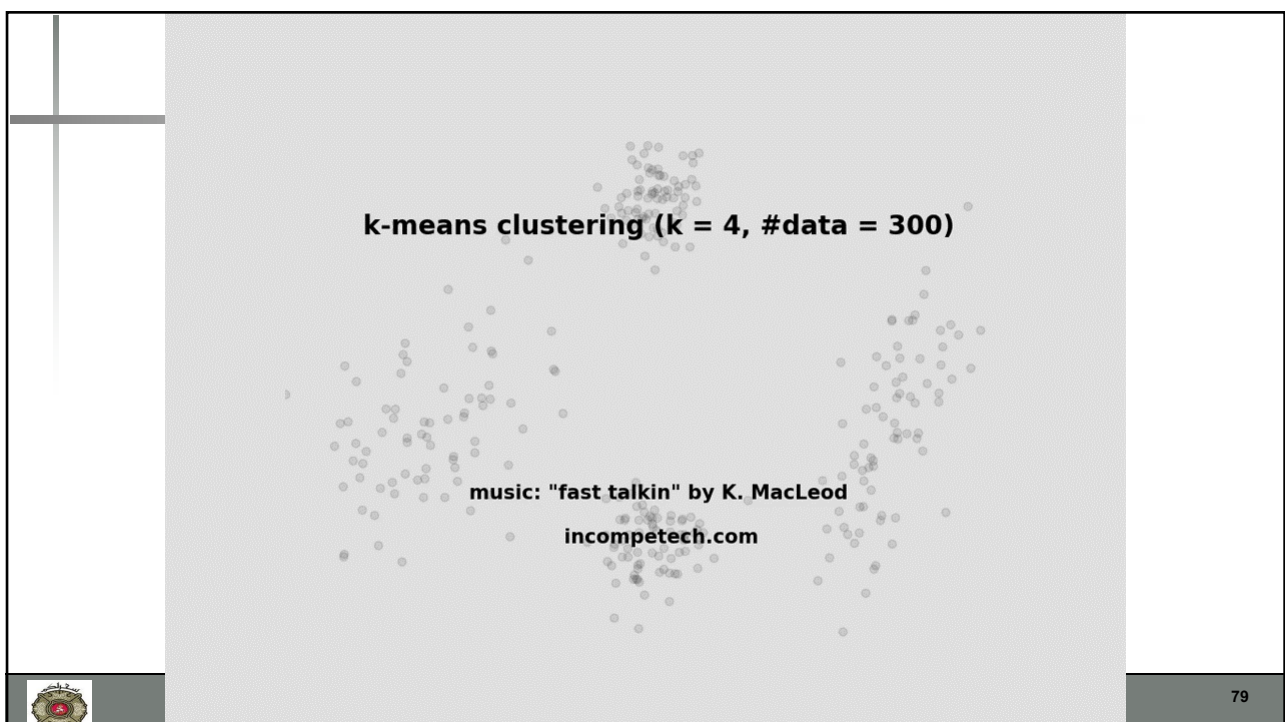
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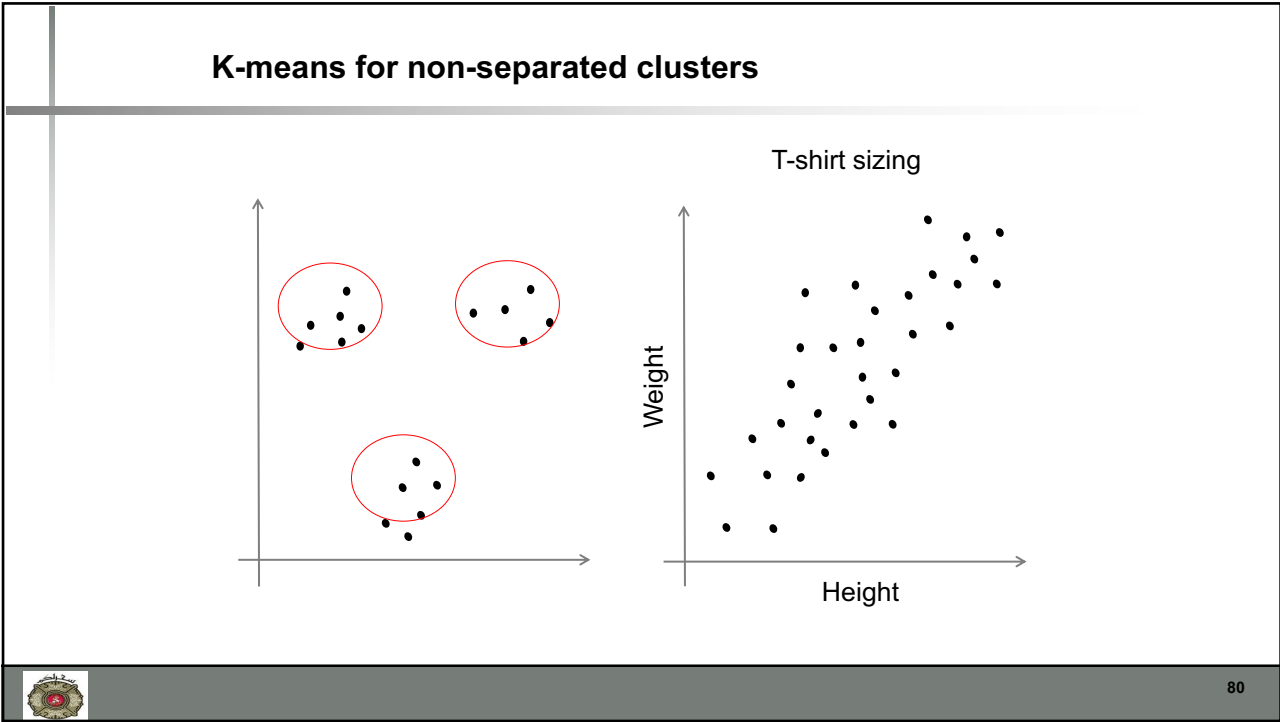
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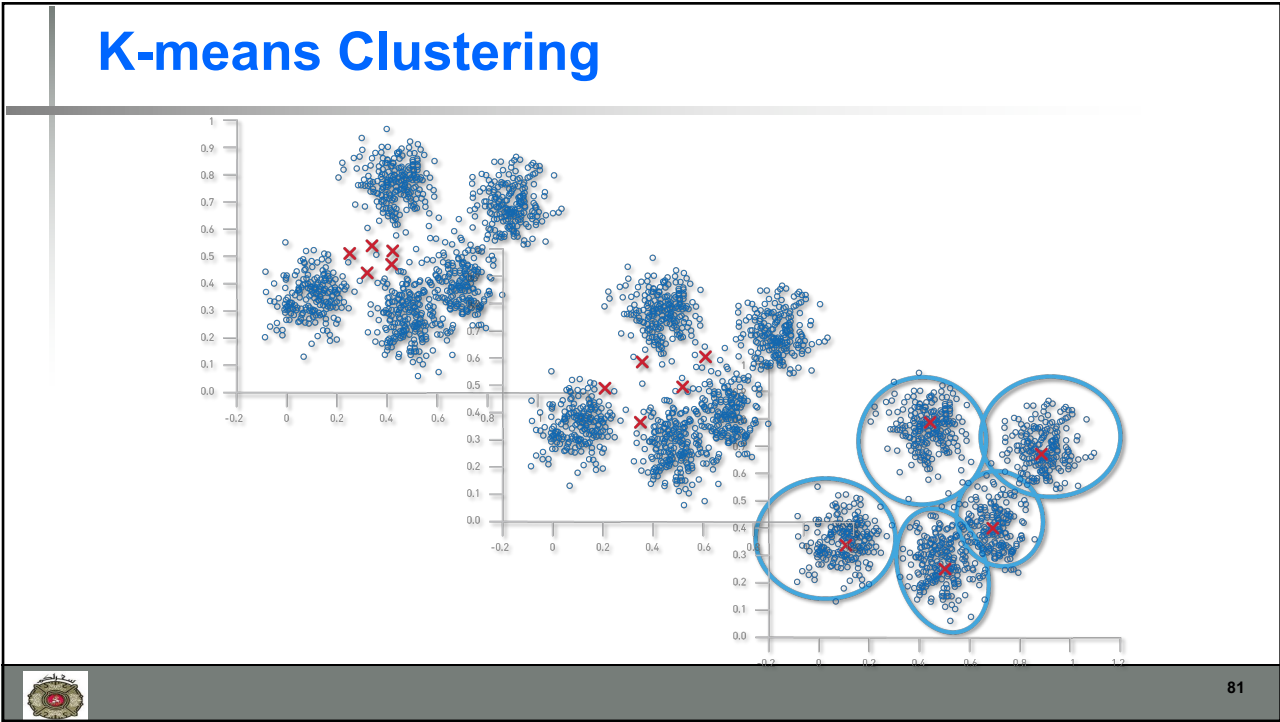
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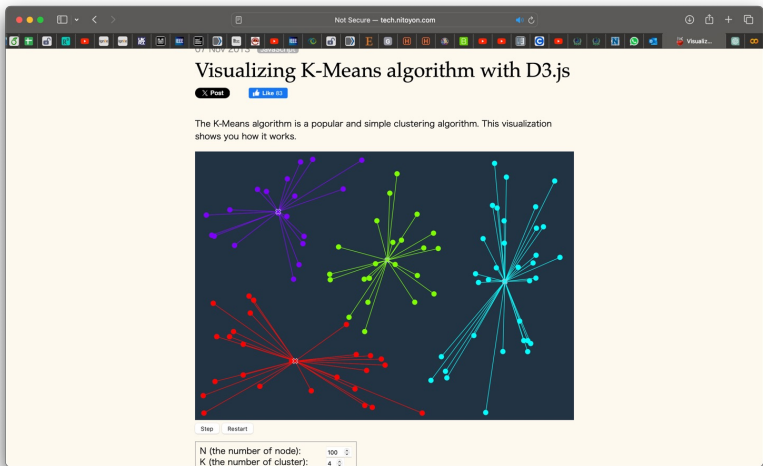


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82

82

K-means clustering



<http://tech.nitoyon.com/en/blog/2013/11/07/k-means/>



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83

83

What distance function should we use?

- Generic form of distance measure
- $dist(x, z) = (\sum_{r=1}^d |x_r - z_r|)^{1/p}$
- Minkowski distance
 - It is a metric in a normed vector space which can be considered as a generalization of both the Euclidean distance and the Manhattan distance.
- $p=1$: Manhattan Distance (l_1 norm)
- $p=2$: Euclidean Distance (l_2 norm)

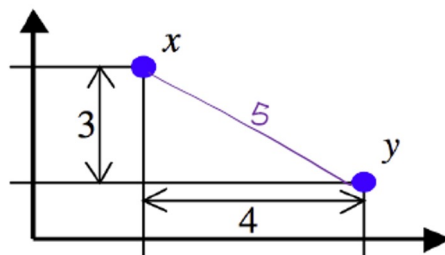


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84

84

Distance Functions for Vectors



- Euclidean distance: $\sqrt{4^2 + 3^2} = 5$
- Manhattan distance: $4 + 3 = 7$
- "inf-distance: $\max\{4, 3\} = 4$



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85

85

OBJECTIVE FUNCTION FOR K MEANS



86

86

K-means optimization objective

$c^{(i)}$ = index of cluster $(1, 2, \dots, K)$ to which example $x^{(i)}$ is currently assigned

μ_k = cluster centroid k ($\mu_k \in \mathbb{R}^n$)

$\mu_{c^{(i)}}$ = cluster centroid of cluster to which example $x^{(i)}$ has been assigned

Optimization objective:

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

$$\min_{c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K} J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$$



87

87

RANDOM INITIALIZATION

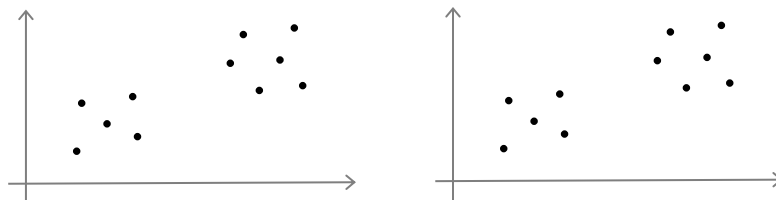


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88

Random initialization

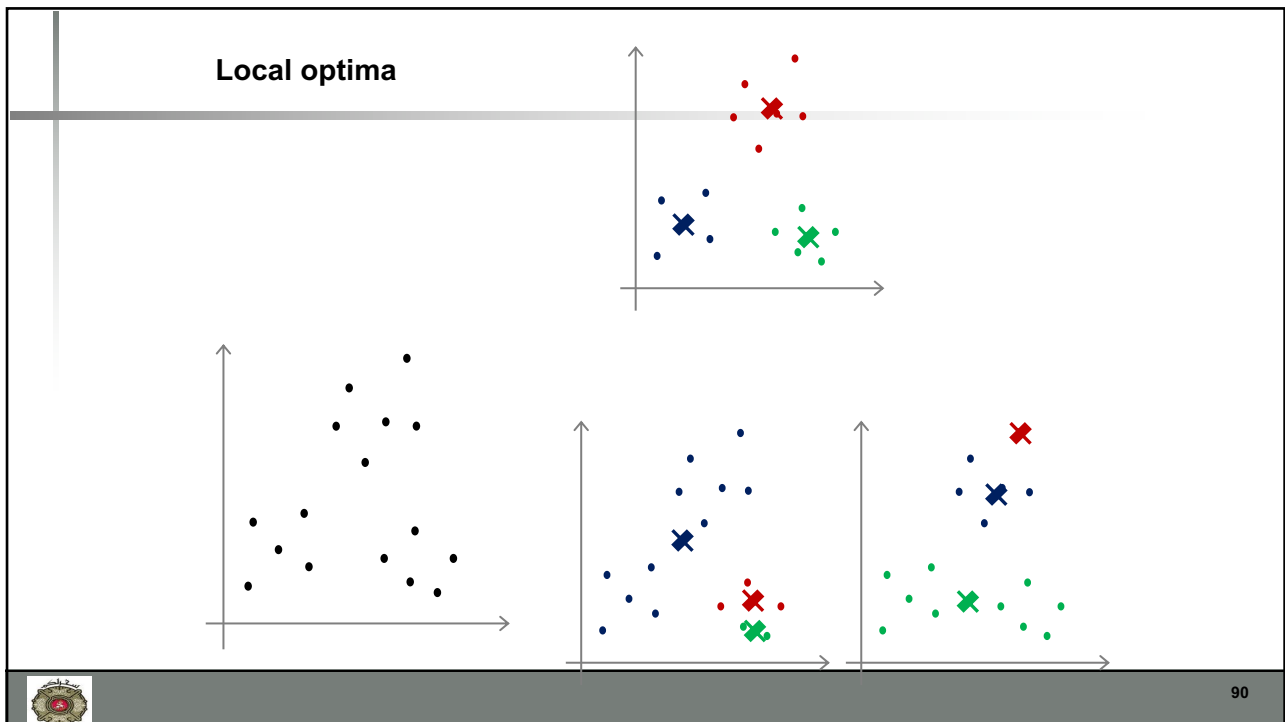
- $K < m$
- Randomly pick K training examples
- Set $\mu_1, \dots, \mu_K$ equal to these K examples



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89

89



90

Random Intialisation

- For $i=1$ to 100 {
Run k-means
}

Pick clusters that gives lowest cost

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$$



A WORKING EXAMPLE



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92

92

Step 1: Choose K

Example:

If you have students' marks in Math and English, and you want to group them into 3 performance levels → set **K = 3**.



93

93

Step 2: Initialize Centroids

Randomly select **K data points** as the initial centroids (cluster centers).

Let's randomly pick **2 centroids**:

- $C_1 = (1, 2)$
- $C_2 = (10, 2)$

Point	x	y
P_1	1	2
P_2	1	4
P_3	1	0
P_4	10	2
P_5	10	4
P_6	10	0



94

94

Step 3: Assign points

Assign points to nearest centroid

Compute its distance (usually Euclidean)

to all centroids and assign it to the nearest one.

Clusters formed:

- **Cluster 1:** P_1, P_2, P_3
- **Cluster 2:** P_4, P_5, P_6

Point	Distance to C_1	Distance to C_2	Nearest
$P_1(1,2)$	0	9	C_1
$P_2(1,4)$	2	9.22	C_1
$P_3(1,0)$	2	9.22	C_1
$P_4(10,2)$	9	0	C_2
$P_5(10,4)$	9.22	2	C_2
$P_6(10,0)$	9.22	2	C_2



95

95

Step 4: Update Centroids

For each cluster, recalculate its centroid as the **mean of all assigned points**:

$$\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$$

New C₁:

$$\left(\frac{1+1+1}{3}, \frac{2+4+0}{3}\right) = (1, 2)$$

New C₂:

$$\left(\frac{10+10+10}{3}, \frac{2+4+0}{3}\right) = (10, 2)$$



96

96

Step 5: Repeat

Repeat **Steps 3–4** until:

- The centroids don't change much, or
- The assignments stop changing, or
- A maximum number of iterations is reached.

Cluster	Points	Centroid
Cluster 1	(1,2), (1,4), (1,0)	(1,2)
Cluster 2	(10,2), (10,4), (10,0)	(10,2)



97

97

Objective

K-Means minimizes the **Within-Cluster Sum of Squares (WCSS)**, also called **inertia**:

$$J = \sum_{j=1}^K \sum_{x_i \in C_j} ||x_i - \mu_j||^2$$

Goal → find cluster assignments and centroids that minimize J.



98

98

Keypoints

- Requires **K** in advance.
- Finds **spherical clusters** (circular in 2D).
- Sensitive to **initialization**.
- Optimizes **within-cluster variance**.



100

100

K MEDOIDS



101

101

Medoid

K-Medoids is a **clustering algorithm** similar to **K-Means**, but instead of using the *mean (average)* of cluster points as the center, it uses an **actual data point** — called the **medoid** — as the cluster representative.



102

102

Robustness

K-Medoids is **more robust to outliers and noise**.

Because:

- The mean (used in K-Means) can be pulled by extreme values.
- The medoid (a real sample) is stable and less affected by outliers.



103

103

Algorithm: Step 1

Initialization

Choose **k** random data points as **initial medoids**.



104

104

Algorithm: Step 2

Assign each data point to the **nearest medoid**, based on some **distance metric** (often Euclidean, Manhattan, etc.).



105

105

Algorithm: Step 3

For each cluster:

- Try swapping the current medoid with a non-medoid point.
- Compute the **total cost (sum of distances)** of all points to their nearest medoid.
- If the swap **reduces total cost**, make the swap (update medoid).



106

106

Algorithm: Step 4

- Repeat assignment and update steps until:
 - Medoids stop changing, or
 - The cost function converges (minimal total distance).



107

107

Cost Function

$$\text{Total Cost} = \sum_{i=1}^k \sum_{x \in C_i} d(x, m_i)$$

- C_i = cluster i
- m_i = medoid of cluster i
- $d(x, m_i)$ = distance between data point and its medoid



108

108

Steps

Choose $k=2$.

1. Randomly select medoids $\rightarrow$ A and D
2. Assign other clusters based on closest medoid:
 - Cluster 1: {A, B}
 - Cluster 2: {C, D, E}
3. Compute total cost (sum of distances)
4. Try swapping A or D with non-medoid points to see if cost decreases.

Continue until no swap improves cost.



109

109

Example # 1

[3, 5, 6, 8, 10, 12, 13]

- $K=2$
- Randomly [3, 13] are initial medoids of cluster 1 and 2
- Distances from 3 = [0, 2, 3, 5, 7, 9, 10]
- Distances from 13 = [10, 8, 7, 5, 3, 1, 0]
- Clusters = [c1, c1, c1, c1, c2, c2, c2]
- Cluster 1 = {3, 5, 6, 8}, Cluster 2 = {10, 12, 13}



110

110

Example

Cluster2 [10, 12, 13]

$$10: (10-12) + (10-13) = 5$$

$$12: (12-10) + (12-13) = 3$$

$$13: (13-10) + (13-12) = 4$$

Next medoid is 12



111

111

Example # 2

Data points: [2, 3, 4, 10, 11, 12]

Suppose $K = 2$

Medoid1 = 3

Medoid2 = 11



112

112

Distances

Point	Distance to 3	Distance to 11	Nearest Medoid	Cluster
2	1	9	3	C1
3	0	8	3	C1
4	1	7	3	C1
10	7	1	11	C2
11	8	0	11	C2
12	9	1	11	C2



113

113

Cluster 1 distances

Candidate	Distances to others	Sum of distances
2	$ 2-3 + 2-4 = 1 + 2$	3
3	$ 3-2 + 3-4 = 1 + 1$	2 ✓
4	$ 4-2 + 4-3 = 2 + 1$	3



Best medoid is 3

114

114

Cluster 2 distances

Candidate	Distances to others	Sum of distances
10	$ 10-11 + 10-12 = 1 + 2$	3
11	$ 11-10 + 11-12 = 1 + 1$	2 ✓
12	$ 12-10 + 12-11 = 2 + 1$	3

Best medoid is 11



115

115

Final result

Cluster	Members	Medoid	In-cluster total distance
C1	[2, 3, 4]	3	2
C2	[10, 11, 12]	11	2



116

116

CHOOSING K



117

117

Clustering

- Clustering: Grouping of objects
- Hard vs. Soft
 - **Hard**: same object can only belong to single cluster
 - e.g. Kmeans clustering
 - **Soft**: same object can belong to different clusters
 - e.g. Gaussian mixture model

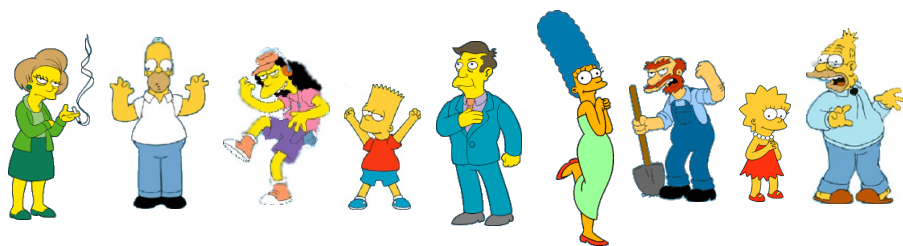


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118

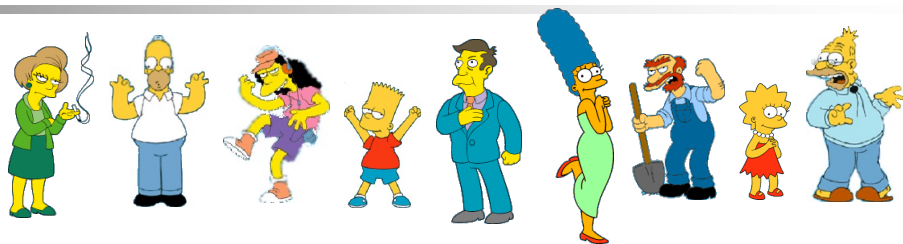
118

Question:
What is a natural grouping among these objects?

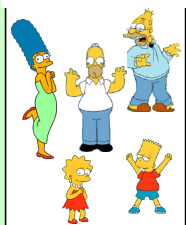


119

What is a natural grouping among these objects?



Clustering is subjective



Simpson's Family



School Employees



Females

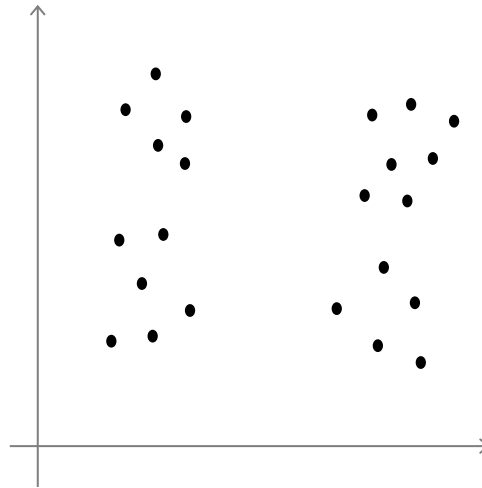


Males



120

What is the right value of K?

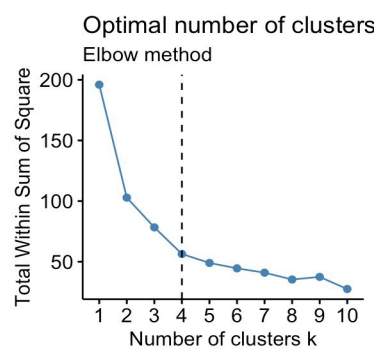


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121

Deciding K

- We can plot objective function values for $k=1$ to 10
- The abrupt change at $k=2$ is highly suggestive of two clusters
- “knee finding” or “elbow finding”
- Note that the results are not always as clear cut as in this toy example



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122

122

ChatGPT Prompt

- can you help me generate python code for kmeans clustering applied to on rgb images for Google CoLab



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123

123

```

KMeanClusteringonImage.py x
1 import cv2
2 import numpy as np
3 from sklearn.cluster import KMeans
4 from google.colab.patches import cv2_imshow # For displaying images in Colab
5
6 # Upload your rib image file to Colab and replace 'your_image.jpg' with the actual filename
7 from google.colab import files
8
9 #uploaded = files.upload()
10
11 # Load the rib image
12 #image_path = next(iter(uploaded))
13 rib_image = cv2.imread('/content/drive/MyDrive/Documents/CEME/Lectures/MachineLearning/Fall2022/Lecture 10 Unsupervised Learning/rib_image.jpg')
14
15 # Reshape the image to a 2D array of pixels (height * width, channels)
16 pixels = rib_image.reshape((-1, 3))
17
18 # Convert pixel values to floating-point
19 pixels = np.float32(pixels)
20
21 # Define the number of clusters (you can adjust this based on your image)
22 num_clusters = 16
23
24 # Apply K-means clustering
25 kmeans = KMeans(n_clusters=num_clusters)
26 kmeans.fit(pixels)
27
28 # Get the cluster centers and labels
29 centers = np.uint8(kmeans.cluster_centers_)
30 labels = kmeans.labels_
31
32 # Assign each pixel to its corresponding cluster center
33 segmented_image = centers[labels.flatten()]
34
35 # Reshape the segmented image back to the original shape
36 segmented_image = segmented_image.reshape(rib_image.shape)
37
38 # Display the original and segmented images in Colab
39 cv2_imshow(rib_image)
40 cv2_imshow(segmented_image)
41
Line 1, Column 1 Tab Size: 4 Python

```



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124

124

Original Image



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125

125

num_clusters = 4



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126

126

num_clusters = 16



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127

127

```

KMeanClustering.py
1 from matplotlib import pyplot as plt
2 from sklearn.cluster import KMeans
3 from sklearn.datasets.samples_generator import make_blobs
4
5 X, y = make_blobs(n_samples=300, cluster_std=1.00, random_state=12)
6 plt.scatter(X[:, 0], X[:, 1])
7 plt.show()
8
9 wcss = []
10 for i in range(1, 11):
11     kmeans = KMeans(n_clusters=i)
12     kmeans.fit(X)
13     wcss.append(kmeans.inertia_)
14 plt.plot(range(1, 11), wcss)
15 plt.title('Elbow Method')
16 plt.xlabel('Number of clusters')
17 plt.ylabel('WCSS')
18 plt.show()
19
20 kmeans = KMeans(n_clusters=7)
21 pred_y = kmeans.fit_predict(X)
22 plt.scatter(X[:, 0], X[:, 1], cmap='viridis')
23 plt.scatter(kmeans.cluster_centers_[0], kmeans.cluster_centers_[1], s=300, c='red')
24 plt.show()

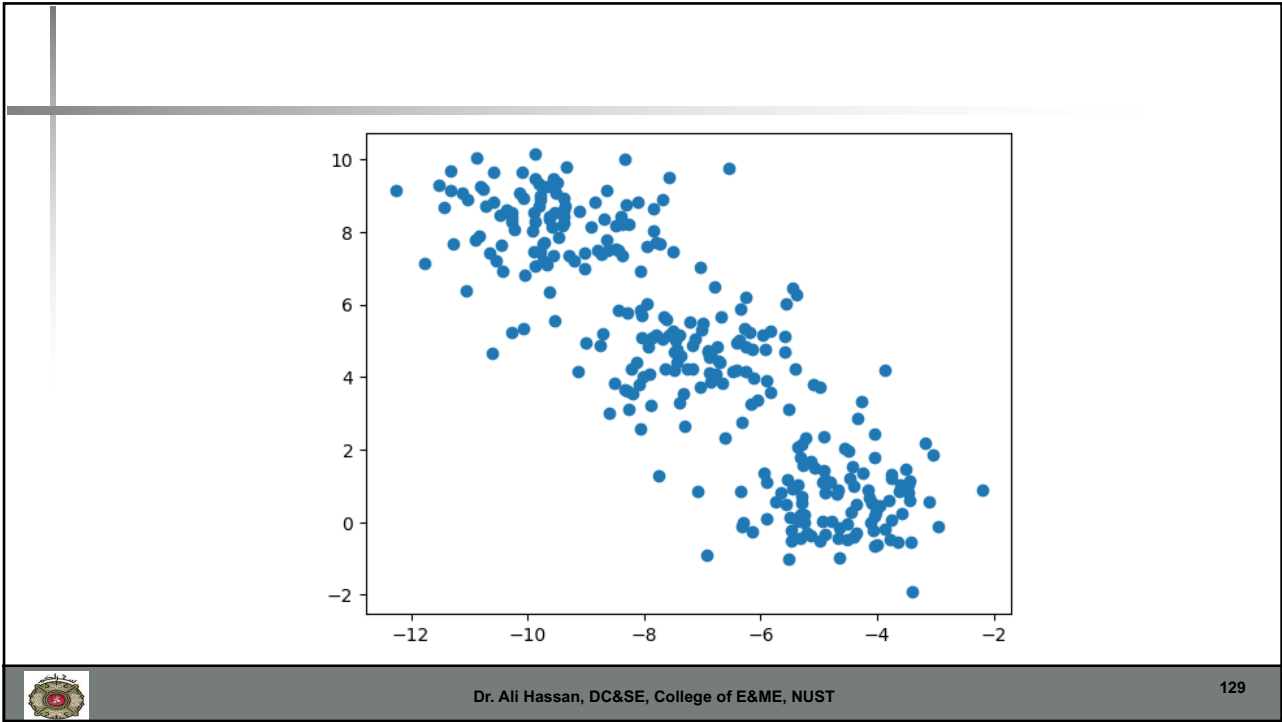
```



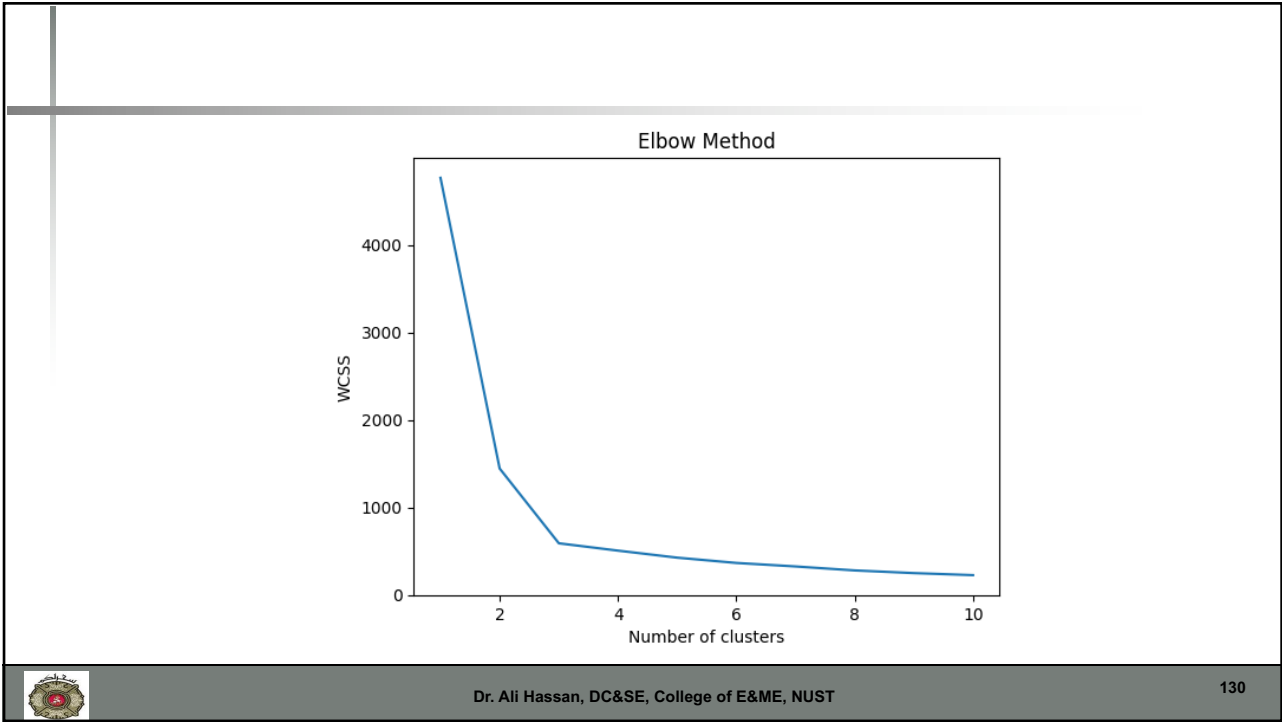
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128

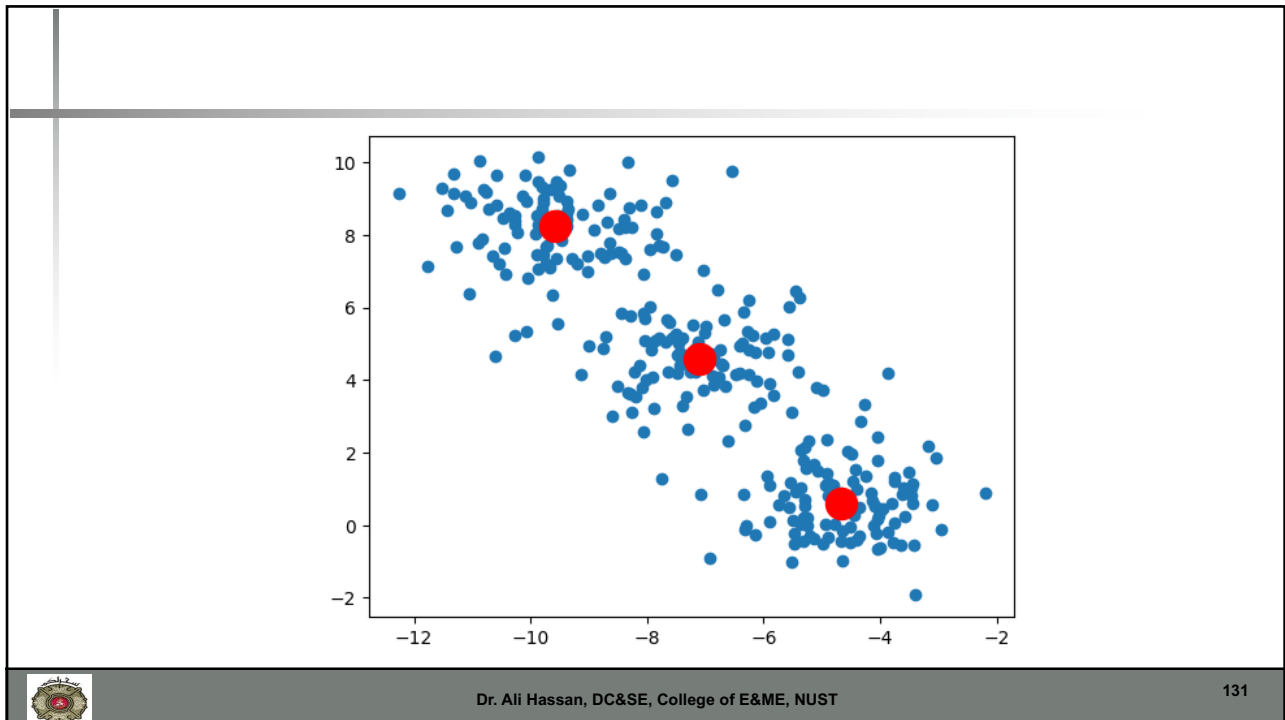
128



129



130



131

ChatGPT Prompt for IRIS Data

- write me a python code to run k-mean clustering on iris data set and plot the elbow method plot for this data for google colab

132

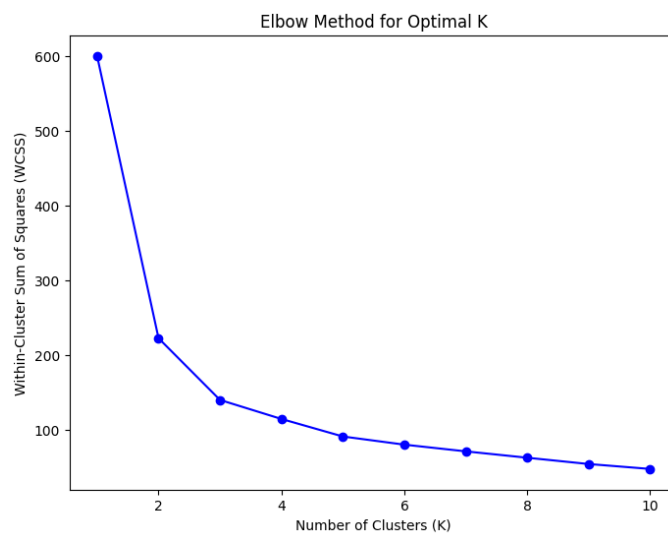
```

1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from sklearn.cluster import KMeans
5 from sklearn import datasets
6 from sklearn.preprocessing import StandardScaler
7 from sklearn.decomposition import PCA
8
9 # Load the Iris dataset
10 iris = datasets.load_iris()
11 data = iris.data
12
13 # Standardize the feature matrix
14 scaler = StandardScaler()
15 data_scaled = scaler.fit_transform(data)
16
17 # Apply PCA to reduce the dimensionality for visualization purposes
18 pca = PCA(n_components=2)
19 data_pca = pca.fit_transform(data_scaled)
20
21 # Determine the optimal number of clusters using the elbow method
22 wcss = [] # Within-Cluster sum of squares
23
24 for i in range(1, 11):
25     kmeans = KMeans(n_clusters=i, init='k-means++', max_iter=300, n_init=10, random_state=0)
26     kmeans.fit(data_scaled)
27     wcss.append(kmeans.inertia_)
28
29 # Plot the elbow method graph
30 plt.figure(figsize=(8, 6))
31 plt.plot(range(1, 11), wcss, markers='o', linestyle='-', color='b')
32 plt.title('Elbow Method for Optimal K')
33 plt.xlabel('Number of Clusters (K)')
34 plt.ylabel('Within-Cluster Sum of Squares (WCSS)')
35 plt.show()
36
37 # Based on the elbow method, choose the optimal number of clusters
38 optimal_k = 3 # You may adjust this based on the plot
39
40 # Perform K-means clustering with the optimal number of clusters
41 kmeans_optimal = KMeans(n_clusters=optimal_k, init='k-means++', max_iter=300, n_init=10, random_state=0)
42 labels_optimal = kmeans_optimal.fit_predict(data_scaled)
43 # Use labels

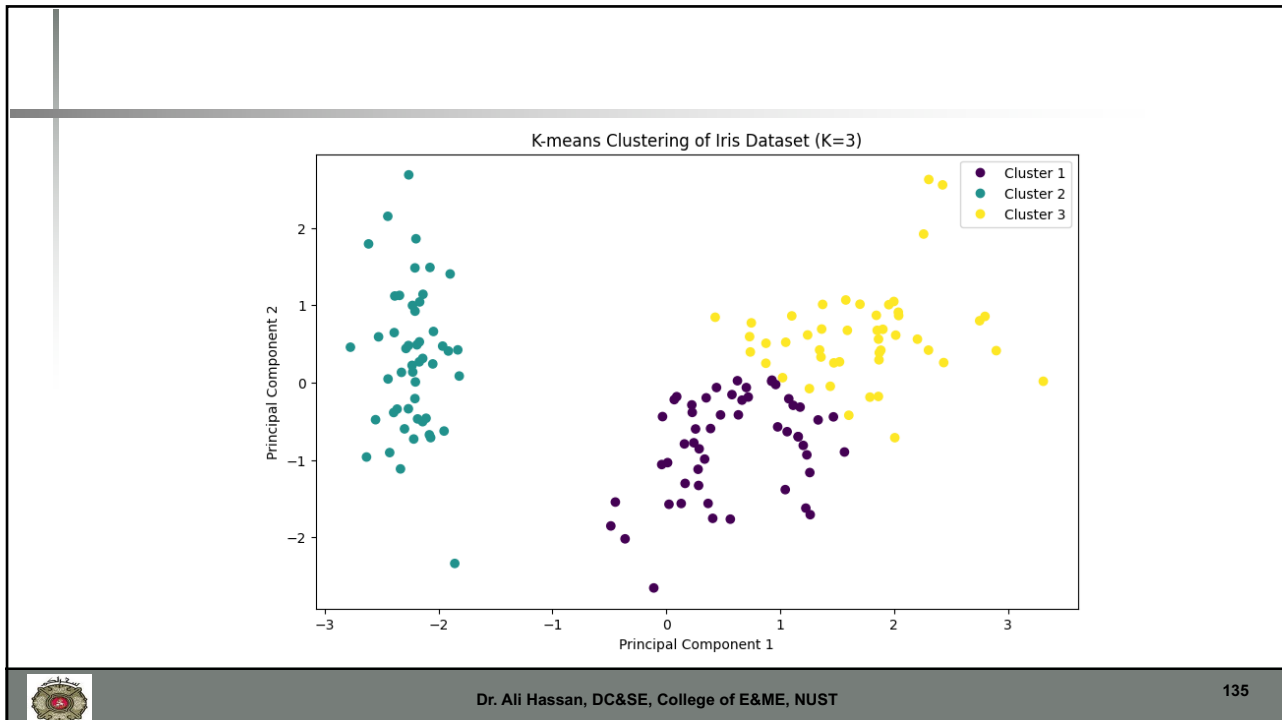
```

133

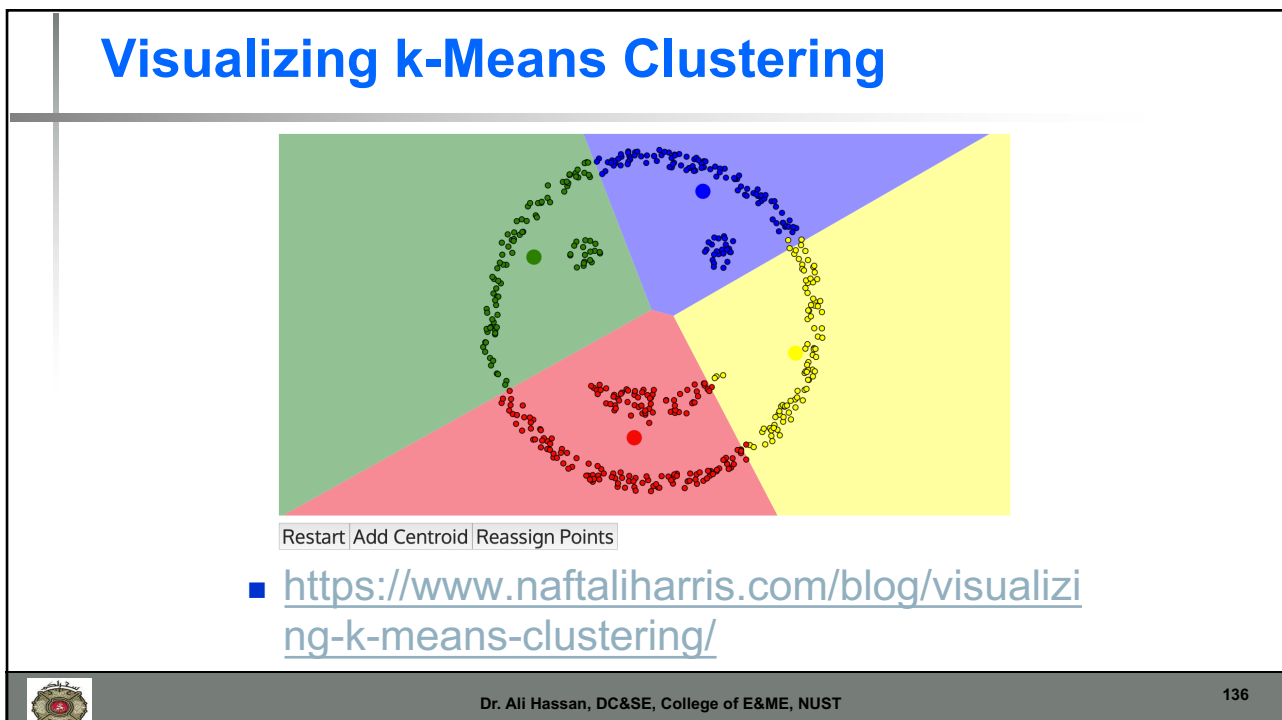
Generated Elbow Plot



134



135



136

Hierarchical Clustering

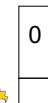
We begin with a distance matrix which contains the distances between every pair of objects in our database.



$$D(\text{Mrs. Krabappel}, \text{Lisa Simpson}) = 8$$



$$D(\text{Marge Simpson}, \text{Bart Simpson}) = 1$$



Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

Consider all possible merges... Choose the best

Consider all possible merges... Choose the best

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139

Starting with each item in its own cluster, find the best pair to merge into a new cluster. Repeat until all clusters are fused together.

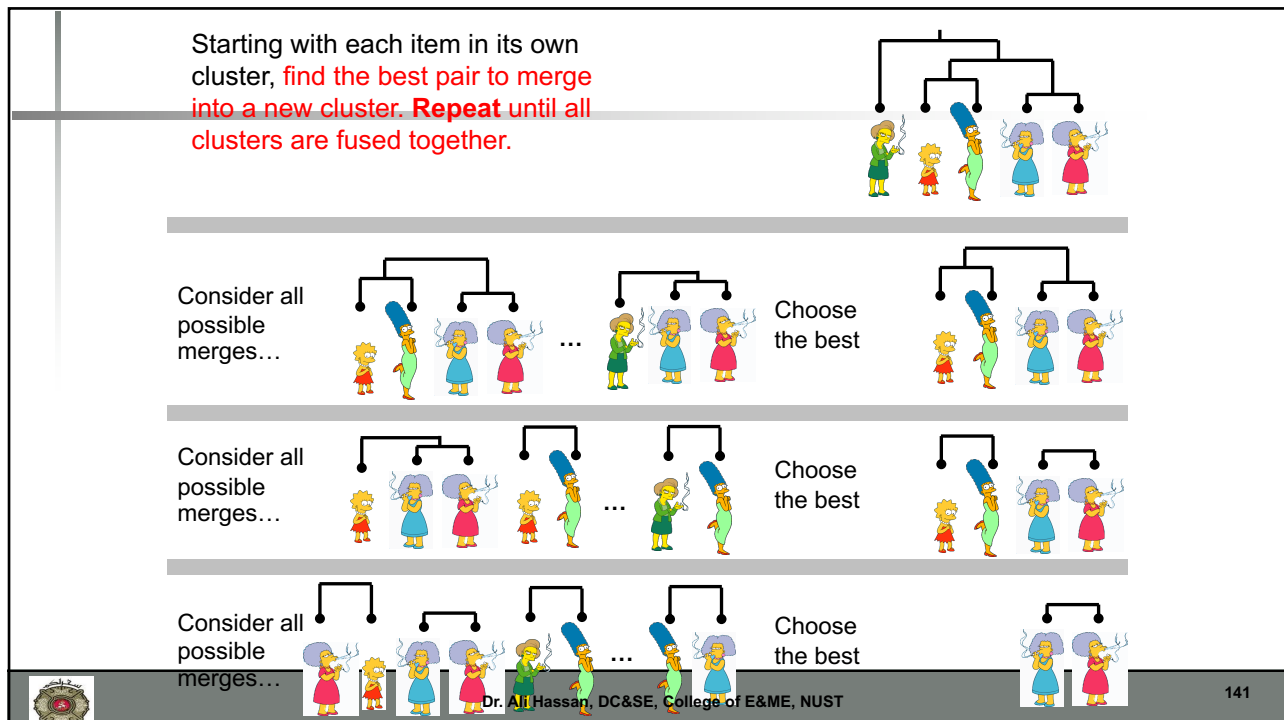
Consider all possible merges... Choose the best

Consider all possible merges... Choose the best

Consider all possible merges... Choose the best

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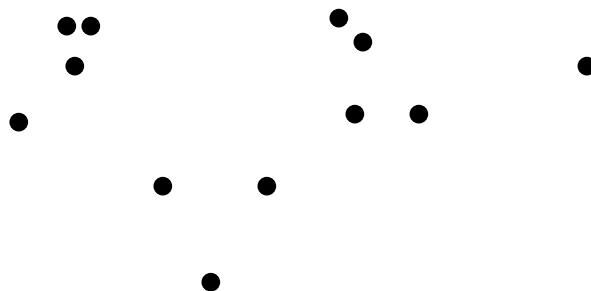
140



141

Hierarchical clustering

■ Bottom up



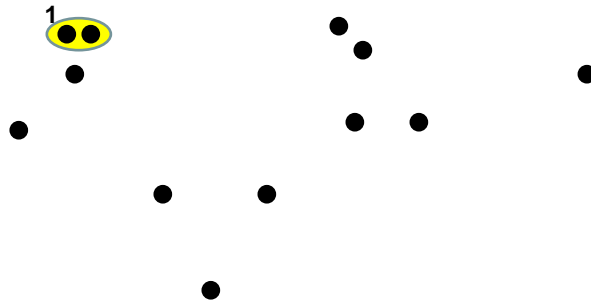
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142

142

Hierarchical clustering

- Bottom up
- 1st iteration



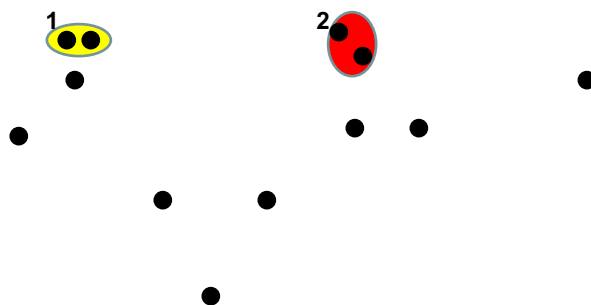
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143

143

Hierarchical clustering

- Bottom up
- 2nd iteration



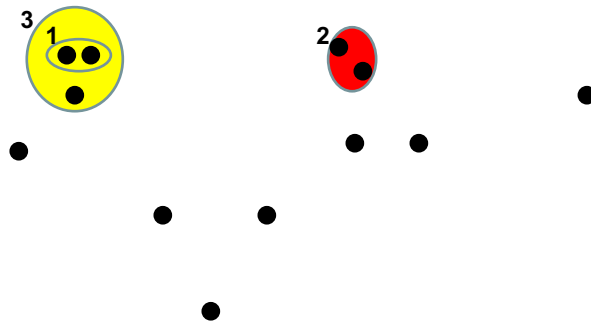
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144

144

Hierarchical clustering

- Bottom up
- 3rd iteration



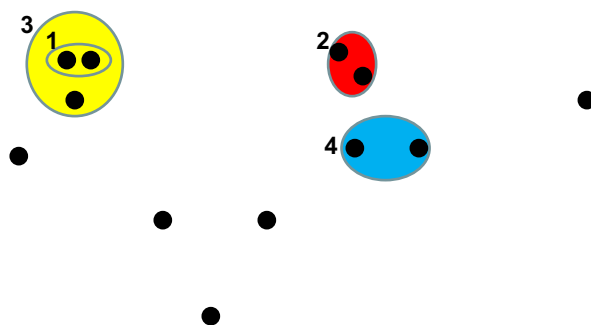
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145

145

Hierarchical clustering

- Bottom up
- 4th iteration



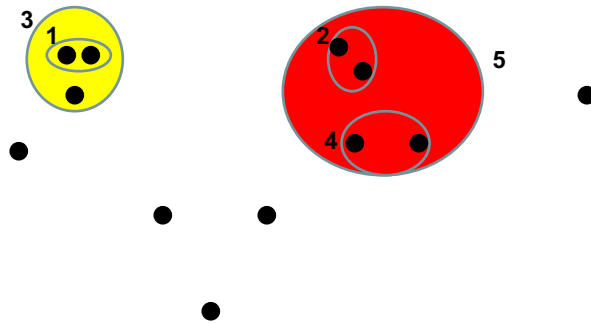
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146

146

Hierarchical clustering

- Bottom up
- 5th iteration



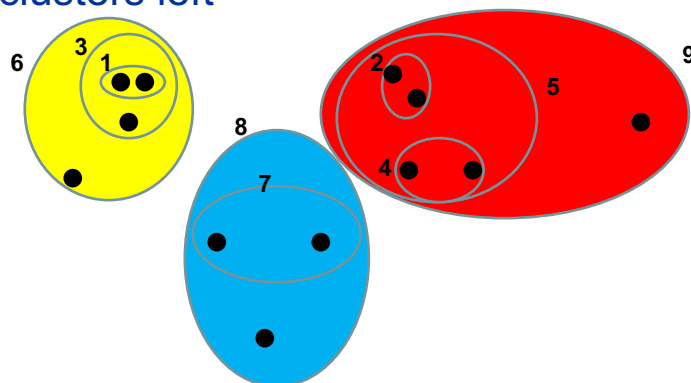
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147

147

Hierarchical clustering

- Bottom up
- Finally k clusters left



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148

148

Summary

■ Strength

- *Relatively efficient*
- Often terminates at *a local optimum*
 - *Run number of times*

■ Weakness

- Unable to handle *noisy data* and *outliers*
- *Expensive (with random seeds)*
- *How to choose initial Seed?*
- Need to *specify k* , the *number* of clusters, in advance
 - Usually driven by application

